



PRINCIPLES OF POINT ESTIMATION

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Point Estimation

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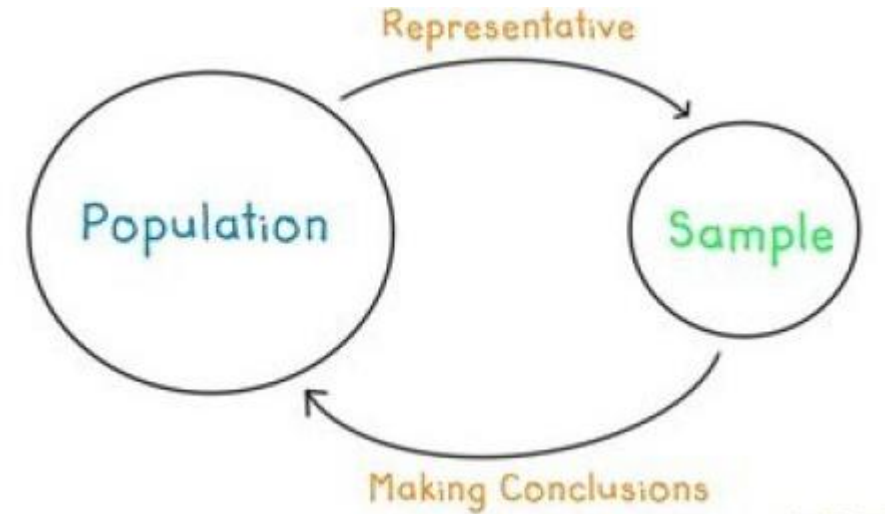
Topics to be covered...



- ✓ Point Estimator and properties .
- ✓ Properties of Point Estimator.
- ✓ Measuring Goodness of an Estimator.
- ✓ Introduction to MSE(Mean Square Error)

- Statistics are used to estimate parameters.
- Descriptive statistics effectively describes the data and it does not make any inference from the data.
- Descriptive refers to the numerical summary of the population which is referred as parameter and for sample it is called as statistic.

- Draw inferences and make conclusion or evaluation about population using the evidence provided by the sample.
- It helps to estimate the parameters of the population.
- The sample may not provide a complete depiction of the population.
- There will always be an uncertainty when drawing conclusions about the population from the sample.

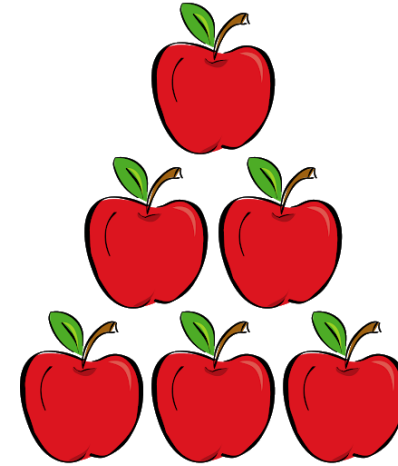


Understanding Point Estimation

- Anna is interested in finding the mean weight of the apples that are imported from Kashmir. However a survey claims that the average weight of an apple is around 90g.

What can we do now?

- It is well known that we cannot weigh every apple (population). So, we need to take samples.
- From those samples we can make inferences about the entire population.
- There are also chances that the samples what we examine will have some errors.



Sampling

- We know that all the apples cannot be weighted, so she decided to take collection of four samples of size 20 each.
- After performing the test, the sample means are displayed below.

Sample 1 : $\bar{x} = 88g$ Sample 3 : $\bar{x} = 91g$

Sample 2 : $\bar{x} = 89.5g$ Sample 4 : $\bar{x} = 89.9g$

What is Point Estimate?

- We have to understand that there is a claim that the average weight of an apple is around 90g.
- From samples we understand that Sample 4 having 89.9g is pretty close to 90g.
- Since this is a sample we cannot expect it be exactly 90g. So, can we claim that these samples accurately reflect the weight parameter of the overall population?
- On the other hand, 89.9g is close enough to 90g and it is plausible to accept. **This is referred as Point Estimate.**

Point Estimate

- It is a single numeric value specified for the data which is also referred as **sample statistic**.
- We collect data for the purpose of estimating some numerical characteristic of the population from which they come.
- A quantity calculated from the data is called a statistic, and a statistic that is used to estimate an unknown constant, or parameter, is called a **point estimator**. Once the data has been collected, we call it a point estimate.
- Point estimate infers about the **population parameters**.

Point Estimators		Inference	Unknown Population Parameters	
Sample Mean	($\bar{X}$)		Population Mean	(μ)
Sample Standard Deviation	(s)		Population Standard Deviation	(σ)
Sample Proportion	($\hat{p}$)		Population Proportion	(p)

Point estimate is used to make an **estimation of unknown population parameters** including population mean, standard deviation and proportion .

Example – Point Estimation of Population Mean

- Point Estimate of Population mean μ is Sample mean $\bar{X}$.
- **Example:** Sample of heights of 34 male freshman students in a class was obtained.

185 161 174 175 202 178 202 139 177
170 151 176 197 214 283 184 189 168
188 170 207 180 167 177 166 231 176
184 179 155 148 180 194 176

$$\bar{X} = 182.44$$

- This can be inferred as the single numeric point estimate for the population mean (true mean) of all the freshman students.

Example – Point Estimation of Population Proportion

- An point estimate of population proportion, p , is the sample proportion

$$\hat{p} = \frac{X}{n}$$

where X is the number of successes in the sample and n is the sample size.

Example: A sample of 100 people were selected in a particular locality to estimate the proportion of them go for walking the park everyday. In this sample 40 do.

$$\hat{p} = \frac{40}{100} = 0.4$$

- For a large population, when sampling technique is used, it is not going to be perfect always. There will always be some uncertainty in estimation.

Property : 1 - Bias

- When the expected value of an estimator is different from the value of the parameter that is being estimated.
- When they are equal, we call it as unbiased.

Property: 2 - Consistency

- This portrays how close the point estimator can be to the true value of the parameter even if it increases in size.
- The consistency and accuracy of point estimator can be achieved by using large samples.
- This can be exercised by mean and the variance.
- To be more consistent the mean of the sample should move towards the true value of the population parameter.

Property: 3 - Efficiency

- A very efficient point estimator should have the following,
 - a) smallest variance.
 - b) unbiased observation.
 - c) consistent.
- All these parameters can be achieved from a normally distributed population.

Introduction to MSE

How can we measure the goodness of an estimator?

- Given a point estimator, how do we determine how good it is?

Goodness Measure - Mean Squared Error(MSE)

- What methods can be used to construct good point estimators?

Good Method to construct Point Estimator - Maximum Likelihood Estimate (MLE)

- A **good estimator** should be both **accurate** and **precise**.
- **Accuracy** of an estimator is measured by **bias**.
- **Precision** is measured by **standard deviation** or **uncertainty**.
- It can be measured by a quantity called **Mean Squared Error (MSE)**.
- **MSE** combines both **bias** and **uncertainty**.

- The **bias** of the estimator is denoted by,

$$\hat{\theta} = \mu_{\hat{\theta}} - \theta$$

- The difference between the mean of the estimator and true value.
- θ is the unknown parameter
- $\hat{\theta}$ denote the estimator of θ
- The **uncertainty** is the standard deviation $\sigma_{\hat{\theta}}$, defined as the standard error of the estimator.

Mean Squared Error (MSE)

- MSE is found by adding the variance to square of the bias.

$$MSE_{\hat{\theta}} = Var(\hat{\theta}) + (Bias\ of\ \hat{\theta})^2$$

- By definition,

Let θ be a parameter, and $\hat{\theta}$ be the estimator of θ .

The mean squared error (MSE) of $\hat{\theta}$ is

$$MSE_{\hat{\theta}} = (\mu_{\hat{\theta}} - \theta)^2 + \sigma_{\hat{\theta}}^2$$

- An equivalent expression is,

$$MSE_{\hat{\theta}} = \mu_{(\hat{\theta} - \theta)^2}$$

Note: $(\hat{\theta} - \theta)^2$ is the difference between estimated value and true value and it is called as error.

Derivation of Mean Squared Error (MSE)

$$\begin{aligned}MSE_{\hat{\theta}} &= E(\hat{\theta} - \theta)^2 \\&= E(\hat{\theta} - \theta)^2 \\&= E(\hat{\theta}^2 + \theta^2 - 2\hat{\theta}\theta) \\&= E(\hat{\theta}^2) + \theta^2 + E(2\hat{\theta}\theta)\end{aligned}$$

Add and subtract by $(E(\hat{\theta}))^2$

$$= (E(\hat{\theta}))^2 + \theta^2 - 2.\theta.E(\hat{\theta}) + E(\hat{\theta}^2) - E(\hat{\theta})^2$$

$$MSE(\hat{\theta}) = (E(\hat{\theta}) - \theta)^2 + E(\hat{\theta}^2) - (E(\hat{\theta}))^2$$

Where:

- First term $(E(\hat{\theta}) - \theta)^2 = (\text{Bias})^2$
- Second term $E(\hat{\theta}^2) - (E(\hat{\theta}))^2 = \text{Variance}$

Problem

Let x, y be independent random variables with:

$$E[x] = E[y] = \mu, \quad \text{Var}(x) = 1, \quad \text{Var}(y) = 4$$

Consider the estimator:

$$T = \frac{2x + y}{3} + 1$$

- a) Find the bias of T .
- b) Compute $\text{Var}(T)$.
- c) Find the MSE of T .

Problem

a) Find the bias of T

$$E[T] = E\left[\frac{2x + y}{3}\right] + 1 = \frac{2\mu + \mu}{3} + 1 = \mu + 1$$

$$\text{Bias} = E[T] - \mu = (\mu + 1) - \mu = 1$$

b) Compute $\text{Var}(T)$

The constant +1 does not affect variance:

$$\begin{aligned}\text{Var}(T) &= \text{Var}\left(\frac{2x + y}{3}\right) \\ &= \frac{4\text{Var}(x) + \text{Var}(y)}{9} = \frac{4 \cdot 1 + 4}{9} = \frac{8}{9}\end{aligned}$$

c) Find the MSE of T

$$\begin{aligned}MSE(T) &= Var(T) + Bias^2 \\&= \frac{8}{9} + (1)^2 = \frac{8}{9} + 1 = \frac{17}{9} \approx 1.89\end{aligned}$$

Problem : Do it Yourself

Let x and y be independent normal random variables with mean μ , $Var(x) = 1$, $Var(y) = 9$.

Consider the estimators:

$$T_1 = \frac{x + 2y}{3}, \quad T_2 = \frac{3x + y}{4}$$

- a) Are T_1 and T_2 unbiased?
- b) Find the variance of each estimator.
- c) Compute the MSE of each estimator.
- d) Which estimator would you prefer at $\mu = 0$? Why?



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MSE for Bernoulli, Binomial and Poisson Distribution:

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Topics to be covered...



- ✓MSE for Bernoulli Distribution.
- ✓MSE for Binomial distribution.
- ✓MSE for Poisson distribution.

Derivation: MSE for Bernoulli with one variable

Define the problem

We have a random variable X with:

$$X \sim \text{Bernoulli}(p).$$

That means:

$$P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

We want to estimate the unknown parameter p based on **only this single observation**.

$$\hat{p} = X.$$

$$\text{Bias}(\hat{p}) = E[\hat{p}] - p.$$

$$E[\hat{p}] = E[X].$$

For a Bernoulli random variable:

$$E[X] = p.$$

So:

$$\text{Bias}(\hat{p}) = \underbrace{p - p}_{=0} = 0.$$

The estimator $\hat{p} = X$ is unbiased.

$$\text{Var}(\hat{p}) = \text{Var}(X).$$

For a Bernoulli random variable:

$$\text{Var}(X) = p(1 - p).$$

The Mean Squared Error is defined by:

$$\text{MSE}(\hat{p}) = E[(\hat{p} - p)^2].$$

There is an important identity:

$$\text{MSE}(\hat{p}) = \text{Var}(\hat{p}) + (\text{Bias}(\hat{p}))^2.$$

Since Bias=0:

$$\text{MSE}(\hat{p}) = \text{Var}(\hat{p}) = p(1 - p).$$

Problem

Let $X \sim \text{Bin}(n, p)$ where p is unknown. Find the MSE of $\hat{p} = X/n$

Solution: As we learnt in the Sample Proportion of Binomial Distribution,

Expectation (to get the bias).

$E[X] = np$ for a $\text{Bin}(n, p)$ variable, so

$$E[\hat{p}] = E\left[\frac{X}{n}\right] = \frac{E[X]}{n} = \frac{np}{n} = p.$$

Hence the bias is

$$\text{Bias}(\hat{p}) = E[\hat{p}] - p = p - p = 0,$$

so $\hat{p}$ is unbiased

Variance.

$\text{Var}(X) = np(1 - p)$ for $X \sim \text{Bin}(n, p)$. Thus

$$\text{Var}(\hat{p}) = \text{Var}\left(\frac{X}{n}\right) = \frac{\text{Var}(X)}{n^2} = \frac{np(1 - p)}{n^2} = \frac{p(1 - p)}{n}.$$

Problem

$$MSE_{\hat{\theta}} = (\text{Bias of } \hat{\theta})^2 + \text{Var}(\hat{\theta}) = 0 + p(1 - p)/n$$

$$MSE_{\hat{\theta}} = p(1 - p)/n$$

Note: When bias is 0. MSE will be equal to variance.

Problem

Determine the MSE of the estimator $\hat{\mu} = \bar{X}$ of the parameter μ of the Poisson(μ) distribution.

Problem

$$X_i \sim Poi(\mu)$$

$$E(x_i) = \mu, \quad Var(x_i) = \mu$$

$$MSE(\hat{\mu}) = Var(\hat{\mu}) + Bias^2(\hat{\mu})$$

$$\begin{aligned} Bias(\hat{\mu}) &= E(\hat{\mu}) - \mu = E(\bar{X}) - \mu = E\left(\frac{1}{n} \sum x_i\right) - \mu \\ &= \frac{1}{n} \sum E(x_i) - \mu = \frac{1}{n} \cdot n\mu - \mu = \mu - \mu = 0 \end{aligned}$$

$$\begin{aligned} Var(\hat{\mu}) &= Var(\bar{X}) = Var\left(\frac{1}{n} \sum x_i\right) = \frac{1}{n^2} \sum Var(x_i) \\ &= \frac{1}{n^2} \cdot n \cdot \mu = \frac{\mu}{n} \end{aligned}$$

Problem

$$MSE(\hat{\mu}) = Var(\hat{\mu}) + Bias^2(\hat{\mu})$$

$$MSE(\hat{\mu}) = \frac{\mu}{n} + o^2$$

$$MSE(\hat{\mu}) = \frac{\mu}{n}$$

Problem

Let $X \sim \text{Binomial}(n = 50, p = 0.4)$.

Define estimator:

$$T = 0.9\hat{p} + 0.05.$$

Find MSE of T .

$$\hat{p} = X/50, \quad E[\hat{p}] = p = 0.4.$$

$$E[T] = 0.9 \times 0.4 + 0.05 = 0.36 + 0.05 = 0.41.$$

True parameter being estimated is $p = 0.4$.

$$\text{Bias} = 0.41 - 0.4 = +0.01.$$

$$\text{Var}(\hat{p}) = \frac{p(1-p)}{n} = \frac{0.4 \times 0.6}{50} = 0.24/50 = 0.0048.$$

$$\text{Var}(T) = (0.9)^2 \times 0.0048 = 0.81 \times 0.0048 = 0.003888.$$

$$\text{MSE} = 0.003888 + (0.01)^2 = 0.003888 + 0.0001 = 0.003988.$$

Problem

$X_1, \dots, X_{30}$ from Poisson with mean $\lambda = 4$.

Estimator:

$$T = \bar{X}^2.$$

Find $E[T]$ and bias.

$\bar{X}$ is unbiased: $E[\bar{X}] = 4$.

$$E[\bar{X}^2] = \text{Var}(\bar{X}) + (E[\bar{X}])^2.$$

$$\text{Var}(\bar{X}) = \frac{\lambda}{n} = 4/30 = 0.1333.$$

$$E[\bar{X}^2] = 0.1333 + 16 = 16.1333.$$

True value we estimate is $\lambda^2 = 16$.

$$\text{Bias} = 16.1333 - 16 = 0.1333.$$

Problem

Let $X \sim \text{Bernoulli}(p = 0.4)$.

Define estimator:

$$T = 0.5X + 0.3.$$

Find MSE of T .

Solution:

$$E[T] = 0.5E[X] + 0.3 = 0.5 \times 0.4 + 0.3 = 0.2 + 0.3 = 0.5.$$

True value $p = 0.4$:

$$\text{Bias} = 0.5 - 0.4 = 0.1.$$

$$\text{Var}(T) = (0.5)^2 \text{Var}(X) = 0.25 \times p(1 - p).$$

$$p(1 - p) = 0.4 \times 0.6 = 0.24.$$

$$\text{Var}(T) = 0.25 \times 0.24 = 0.06.$$

$$\text{MSE} = 0.06 + (0.1)^2 = 0.06 + 0.01 = 0.07.$$

Problem: Do it itself.

$X \sim \text{Bernoulli}(p = 0.6)$.

Define estimator:

$$T = 1 - X.$$

Compute MSE of T .

$X \sim \text{Binomial}(n = 100, p = 0.3)$.

Estimator:

$$T = 0.85\hat{p} + 0.05.$$

Compute MSE.

$X_1, \dots, X_{10}$ from Poisson with mean $\lambda = 3$.

Define estimator:

$$\tilde{\lambda} = \bar{X} + 0.5.$$

Find MSE of $\tilde{\lambda}$.



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MSE for Normal Distribution:

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Topics to be covered...

✓ MSE for Normal Distribution.

Problem

Determine the MSE of the estimator $\hat{\mu} = \bar{X}$ of the parameter μ of the normal(μ) distribution.

Problem

The MSE of an estimator $\hat{\theta}$ for a parameter θ is given by:

$$\text{MSE}(\hat{\theta}) = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta})$$

For the sample mean $\hat{\mu}$, we compute its **bias** and **variance**.

The **bias** of $\hat{\mu}$ is:

$$\text{Bias}(\hat{\mu}) = \mathbb{E}[\hat{\mu}] - \mu$$

Since $\mathbb{E}[\hat{\mu}] = \mu$ (the sample mean is unbiased), we have:

$$\text{Bias}(\hat{\mu}) = 0$$

Problem

The variance of $\hat{\mu}$ is:

$$\text{Var}(\hat{\mu}) = \text{Var} \left(\frac{1}{n} \sum_{i=1}^n X_i \right)$$

Because the X_i are **independent**, the variance of the sum is the sum of the variances:

$$\text{Var}(\hat{\mu}) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i)$$

Since $\text{Var}(X_i) = \sigma^2$ for all i :

$$\text{Var}(\hat{\mu}) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}$$

Since the bias is zero:

$$\text{MSE}(\hat{\mu}) = \text{Bias}(\hat{\mu})^2 + \text{Var}(\hat{\mu}) = 0 + \frac{\sigma^2}{n} = \boxed{\frac{\sigma^2}{n}}$$

Problem

Let $X \sim N(\theta, 4)$, where θ is unknown. Consider the estimator $\hat{\theta} = aX + b$, where a and b are constants.

- (a) Find the bias of $\hat{\theta}$.
- (b) Find the variance of $\hat{\theta}$.
- (c) Calculate the MSE of $\hat{\theta}$.
- (d) For $\theta = 3$, $a = 0.5$, and $b = 1$, compute the numerical value of the MSE.

Problem

Given:

- $X \sim N(\theta, 4)$ (so $\text{Var}(X) = 4$).
- Estimator: $\hat{\theta} = aX + b$.

(a) Bias of $\hat{\theta}$

The bias is $\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$.

$$E[\hat{\theta}] = E[aX + b] = aE[X] + b = a\theta + b$$

$$\text{Bias}(\hat{\theta}) = (a\theta + b) - \theta = (a - 1)\theta + b$$

(b) Variance of $\hat{\theta}$

$$\text{Var}(\hat{\theta}) = \text{Var}(aX + b) = a^2 \text{Var}(X) = a^2 \cdot 4 = 4a^2$$

Problem

MSE of $\hat{\theta}$

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + \text{Bias}^2 = 4a^2 + [(a - 1)\theta + b]^2$$

Numerical MSE for $\theta = 3, a = 0.5, b = 1$

Substitute into the MSE formula:

$$\text{MSE}(\hat{\theta}) = 4(0.5)^2 + [(0.5 - 1)(3) + 1]^2 = 4(0.25) + (-1.5 + 1)^2 = 1 + (-0.5)^2 = 1 + 0.25$$

Problem

Let $X \sim N(\theta, 1)$. Define the estimator $\hat{\theta}_6 = X + c$, where c is a constant.

- (a) Find the bias of $\hat{\theta}_6$.
- (b) Calculate its variance.
- (c) Show that the MSE is $1 + c^2$.
- (d) For what value of c does the MSE equal 2?

(a) Find the bias of $\hat{\theta}_6$.

The bias of an estimator $\hat{\theta}$ is defined as:

$$\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$$

Given $\hat{\theta}_6 = X + c$, and $X \sim N(\theta, 1)$, so $E[X] = \theta$.

$$E[\hat{\theta}_6] = E[X + c] = E[X] + c = \theta + c$$

Therefore,

$$\text{Bias}(\hat{\theta}_6) = (\theta + c) - \theta = c$$

Problem

(b) Calculate the variance of $\hat{\theta}_6$.

Since c is a constant, variance is unaffected by adding a constant:

$$\text{Var}(\hat{\theta}_6) = \text{Var}(X + c) = \text{Var}(X) = 1$$

© Show that the Mean Squared Error (MSE) is $1 + c^2$.

Recall:

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

Using the results from (a) and (b):

$$\text{MSE}(\hat{\theta}_6) = 1 + c^2$$

(d) For what value of c does the MSE equal 2?

Set:

$$1 + c^2 = 2 \implies c^2 = 1 \implies c = \pm 1$$

Problem

Let $X_1, \dots, X_n \sim N(\mu, \sigma^2)$. Consider the estimator $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$. Derive the MSE of $\hat{\sigma}^2$.

The MSE is:

$$\text{MSE}(\hat{\sigma}^2) = \text{Var}(\hat{\sigma}^2) + (\text{Bias}(\hat{\sigma}^2))^2$$

- Bias:

$$\text{Bias}(\hat{\sigma}^2) = E[\hat{\sigma}^2] - \sigma^2 = \frac{n-1}{n} \sigma^2 - \sigma^2 = -\frac{\sigma^2}{n}$$

- Variance of $\hat{\sigma}^2$ (for normal samples):

$$\text{Var}(\hat{\sigma}^2) = \frac{2(n-1)}{n^2} \sigma^4$$

Therefore,

$$\text{MSE}(\hat{\sigma}^2) = \frac{2(n-1)}{n^2} \sigma^4 + \left(-\frac{\sigma^2}{n}\right)^2 = \frac{2(n-1)}{n^2} \sigma^4 + \frac{\sigma^4}{n^2} = \frac{2(n-1) + 1}{n^2} \sigma^4 = \frac{2n-1}{n^2} \sigma^4$$

Problem: Do it Yourself

Let $X \sim N(\theta, 1)$. Define $\hat{\theta}_2 = \alpha X + (1 - \alpha)c$, where c is a known constant.

- (a) Find the bias and show it is $\alpha\theta + (1 - \alpha)c - \theta$.
- (b) Prove the variance is α^2 .

Let $X_1, X_2 \sim N(\theta, 1)$ be independent. Consider the estimator $\hat{\theta} = kX_1 + (1 - k)X_2$.

- (a) For what value of k is $\hat{\theta}$ unbiased?
- (b) Find the variance of $\hat{\theta}$ as a function of k .
- (c) Find the value of k that minimizes the MSE.



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Maximum Likelihood Estimation

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Maximum Likelihood Estimation



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Topics to be covered...

✓The Method of Maximum Likelihood for Bernoulli,
Binomial and Poisson Distributions.



Maximum Likelihood Estimate (MLE)

- If we identify a population following any distribution, but the population (parameters) mean and variance are unknown.
- By taking adequate number of samples from the population, by finding the mean and variance so that the observed data is the one which is most likely to occur.
- Maximum Likelihood Estimate (MLE) is the good method that can be applied for estimating parameters.
- It can be obtained from any given distribution using the observed data.
- The suggestion is to estimate the parameter with the value that makes the observed data most likely.

Definition



Let $X_1, \dots, X_n$ have joint probability density or probability mass function

$f(x_1, \dots, x_n; \theta_1, \dots, \theta_k)$, where $\theta_1, \dots, \theta_k$ are parameters and

$x_1, \dots, x_n$ are values observed from $X_1, \dots, X_n$.

The values $\hat{\theta}_1, \dots, \hat{\theta}_k$ that maximize f are the
maximum likelihood estimates of $\theta_1, \dots, \theta_k$

If the random variables $X_1, \dots, X_n$ are substituted for $x_1, \dots, x_n$, then

$\hat{\theta}_1, \dots, \hat{\theta}_k$ are called maximum likelihood estimators

The abbreviation MLE is often used for both maximum likelihood estimate
and maximum likelihood estimator

Points to Remember



- The **maximum likelihood estimate** is the value of the estimators that when substituted in for the parameters maximizes the likelihood function.
- The likelihood function can be a probability density function or a probability mass function.
- It can also be a joint probability density or mass function, and is often the joint density or mass function of independent random variables.
- **Note: Joint probability is the probability of occurrence of two or more events together.**

Obesity in Women – Understanding MLE

Let's start with an example. Annie is a post graduate student who want to study on the growing health problems with women due to obesity. She decided to collect data from the samples she had chosen between age of 20 – 25 years. She surveys with the questionnaire of the diet and exercise habits of her 10 class mates to start with and collects their weights and plots it from low to high.

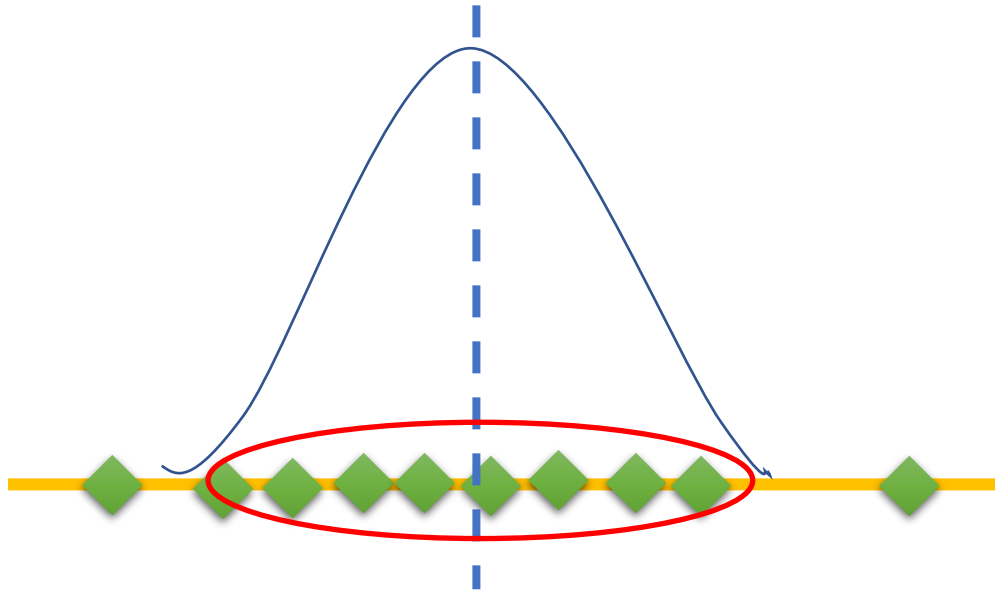
And it looks like this,



From the collected sample estimates, she intends to make inferences for the population parameters.

Obesity in Women – MLE for Normal Distribution

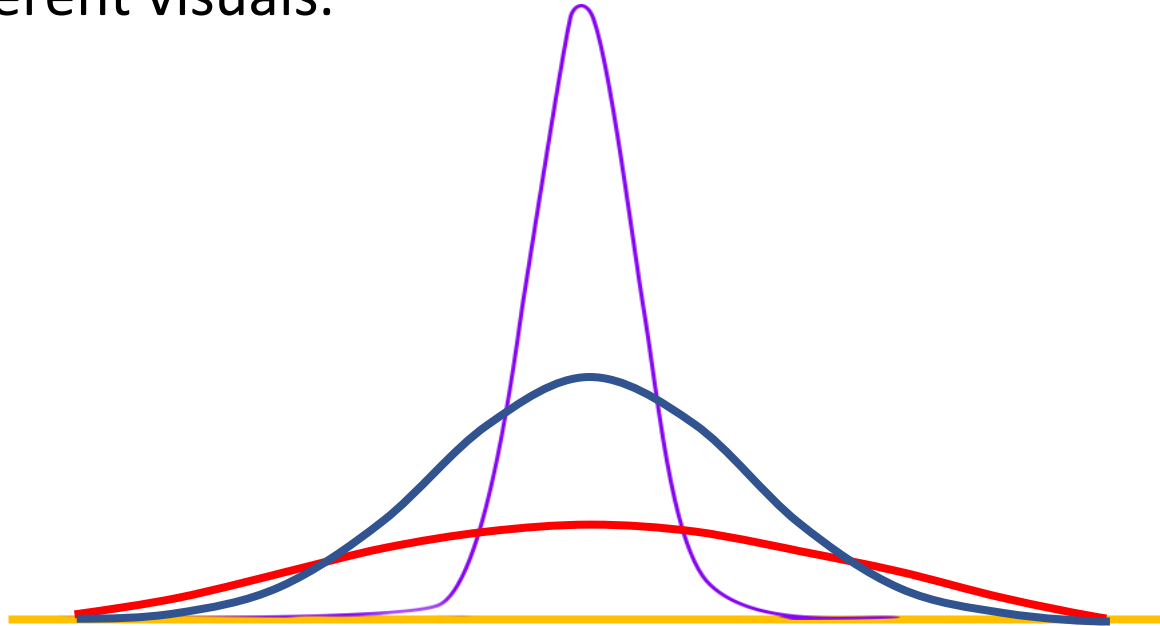
At first step, she has to decide on which model can describe her data best. From the plot she sees that weights are adequately described by the Normal distribution.



We can understand that from the plot seen, it suggests that Normal distribution is plausible because most of the data points are clustered around the middle and few scattered to the left and right.

Obesity in Women – How do we go about the distribution?

We already know that Normal distribution has two parameters associated with it. The mean μ and standard deviation σ . Different values for these parameters may give us different visuals.



MLE is the method that would help us in finding the value of μ and σ that will result in the bell curve that fits our data best in.

Step 1: Write down the likelihood function.

Step 2: Take natural log of likelihood function.

(Reason: the quantity that maximizes log of a function is always the same quantity that maximizes the function itself)

Step 3: Differentiate log-likelihood function with respect to the parameter being estimated.

Step 4: Set the derivative equal to 0 to get MLE.

Bernoulli Distribution – Estimate Likelihood Function for (p)

Let $X_1, \dots, X_n$ be a random sample from the population with Bernoulli (p) distribution.

Find MLE of p.

A random variable X follows a Bernoulli distribution if:

$$X \sim \text{Bernoulli}(p), \quad \text{where } P(X = 1) = p \text{ and } P(X = 0) = 1 - p.$$

The probability mass function (PMF) is:

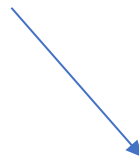
$$P(X = x) = p^x (1 - p)^{1-x}, \quad x \in \{0, 1\}.$$

For n observations,

$$f(x_1, x_2, \dots, x_n / p) = p^{x_1} (1 - p)^{1-x_1} \times p^{x_2} (1 - p)^{1-x_2} \times \dots \times p^{x_n} (1 - p)^{1-x_n}$$

=product of individual density functions

$$P(X_1=x_1, X_2=x_2, \dots, X_n=x_n / p) = \prod_{i=1}^n p^{x_i} (1 - p)^{1-x_i}$$



Joint Probability Mass/Density
(distribution) function

$$P(X_1=x_1, X_2=x_2, \dots, X_n=x_n/p) = f(x_1:p) f(x_2:p) \dots f(x_n:p)$$

$$= \prod_{i=1}^n f(x_i; p)$$

$$= \prod_{i=1}^n p^{x_i} (1 - p)^{1-x_i}$$

$$= L(p)$$

= Likelihood Function

for $0 < p < 1$. Simplifying, by summing up the exponents, we get :

$$L(p) = p^{\sum x_i} (1 - p)^{n - \sum x_i}$$

Step 1: Write the log-likelihood function

Taking the natural logarithm:

$$\ell(p) = \log L(p) = \sum_{i=1}^n [x_i \log p + (1 - x_i) \log(1 - p)].$$

Step 2: Differentiate the log-likelihood with respect to p

$$\frac{d\ell}{dp} = \sum_{i=1}^n \left[\frac{x_i}{p} - \frac{1 - x_i}{1 - p} \right] = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1 - p}.$$

Step 3: Set derivative equal to zero to find critical points

$$\frac{\sum x_i}{p} - \frac{n - \sum x_i}{1 - p} = 0.$$

Multiply both sides by $p(1 - p)$:

$$\sum x_i(1 - p) - (n - \sum x_i)p = 0.$$

Step 4: Simplify and solve for p

$$\sum x_i - p \sum x_i - np + p \sum x_i = \sum x_i - np = 0.$$

Therefore,

$$np = \sum x_i \implies \hat{p} = \frac{1}{n} \sum_{i=1}^n x_i.$$

$$p = \frac{\sum x_i}{n} = \bar{X} \quad \text{The MLE of } p \text{ is } \hat{p} = \bar{X}$$

Example – Binomial Distribution – Estimate Likelihood Function



Consider the following example,

$X \sim \text{Bin}(20, p)$, p is unknown

Suppose we observe $X = 7$. The probability mass function is,

$$f(7; p) = \frac{20!}{7!13!} p^7 (1 - p)^{13}$$

In the probability mass function it is as written $f(7; p)$ rather than $f(7)$. Here the data value 7 is constant.

When a probability mass function or probability density function is considered to be a function of parameters, it is called a **likelihood function**.

How can we maximise this likelihood function?

When $\hat{p}$ is substituted for p , it maximizes the likelihood function

Let's compute maximum likelihood function for $f(7; p)$

We could maximize this function by taking the derivative with respect to p and setting it equal to 0.

$$\ln f(7; p) = \ln 20! - \ln 7! - \ln 13! + 7 \ln p + 13 \ln(1 - p)$$

We take the derivative with respect to p and set it equal to 0.

$$\frac{d}{dp} \ln f(7; p) = \frac{7}{p} - \frac{13}{1 - p} = 0$$

The maximizing value is $\frac{7}{20}$

Therefore the maximum likelihood estimate is $\hat{p} = \frac{7}{20}$

Let $X \sim \text{Bin}(n, p)$ where n is known and p is unknown

Find MLE of p .

The probability mass function of X is $f(x; n, p) = \frac{n!}{x! (n-x)!} p^x (1-p)^{n-x}$

The likelihood function is given by,

$$L(p; n, x) = \frac{n!}{x! (n-x)!} p^x (1-p)^{n-x}$$

Take log of the likelihood function,

$$\begin{aligned} \ln L(p; n, x) &= \ln \left[\frac{n!}{x! (n-x)!} p^x (1-p)^{n-x} \right] \\ &= \ln \left(\frac{n!}{x! (n-x)!} \right) + x \ln p + (n-x) \ln(1-p) \end{aligned}$$

Taking derivative with respect to p and set it to 0,

$$\begin{aligned}\frac{d}{dp} L(p; n, x) &= \frac{d}{dp} (\ln n! - \ln x! - \ln(n-x)! + x \ln p + (n-x) \ln(1-p)) \\ &= 0 + \frac{x}{p} - \frac{x}{1-p} - \frac{n}{1-p}\end{aligned}$$

Apply chain rule,

Multiply both sides by $p(1-p)$

$$0 = x - xp + xp - np$$

$$p = \frac{x}{n}$$

The MLE of p is $\hat{p} = \frac{X}{n}$

Let $X \sim \text{Bin}(n, p)$ where n is known and p is unknown

Find the MLE of the odds ratio $p/(1 - p)$

The MLE of p is $\hat{p} = \frac{X}{n}$

The MLE of $\frac{p}{1-p} = \frac{\hat{p}}{1-\hat{p}} = \frac{X}{n-X}$

Let $X_1, \dots, X_n$ be an independent random sample from a population with $\text{poisson}(\lambda)$. Find MLE of λ .

The probability mass function is given as,

$$L(x_1, \dots, x_n | \lambda) = e^{-\lambda} \frac{\lambda^{x_i}}{x_i!}$$

The likelihood function

$$L(x_1, \dots, x_n; \lambda) = \frac{\lambda^{\sum x_i} e^{-n\lambda}}{\prod_{i=1}^n x_i!}$$

This is the log of likelihood function

$$\ln L(x_i; \lambda) = \sum x_i \ln(\lambda) - n\lambda \ln(e) - \ln(x_1!, \dots, x_n!)$$

The MLE is the value of λ that maximizes $f(x_1, \dots, x_n; \lambda)$, or equivalently, $\ln f(x_1, \dots, x_n; \lambda)$.

$$\frac{d}{d\lambda} \ln f(x_1, \dots, x_n; \lambda) = \frac{d}{d\lambda} (-n\lambda + \sum_{i=1}^n x_i \ln \lambda - \sum_{i=1}^n \ln x_i!) = -n + \frac{\sum_{i=1}^n x_i}{\lambda} = 0.$$

Now we solve for λ :

$$-n + \frac{\sum_{i=1}^n x_i}{\lambda} = 0.$$

$$\lambda = \frac{\sum_{i=1}^n x_i}{n} = \bar{x}.$$

The MLE is $\hat{\lambda} = \bar{X}$.

Example - Poisson Distribution

Problem:

The following data are the observed frequencies of occurrence of domestic accidents: we have $n = 647$ data as follows'

Number of Accidents	Frequency
0	447
1	132
2	42
3	21
4	3
5	2

What is the estimate of λ if a Poisson model is assumed ?

Example - Poisson Distribution

Solution:

The MLE of λ is $\hat{\lambda} = \bar{X}$

$$\begin{aligned}\hat{\lambda} &= \frac{1}{n} \sum_{i=1}^n x_i = \bar{X} \\ &= \frac{(447 * 0) + (132 * 1) + (42 * 2) + (21 * 3) + (3 * 4) + (2 * 5)}{674} \\ &= 0.465\end{aligned}$$



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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Maximum Likelihood Estimation

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Maximum Likelihood Estimation



MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Topics to be covered...

✓ The Method of Maximum Likelihood for Normal Distribution.

✓ Pitfalls of Point Estimators

Let $x_1, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$ population. Find MLE of μ and σ .

The probability of observing data points that can be generated by Normal distribution is,

$$f(x_i; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

The joint probability function of $X_1, \dots, X_n$ is,

$$L(x_1, \dots, x_n; \mu, \sigma) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

for $-\infty < x < \infty$. The parameter space is $\Omega = \{(\mu, \sigma): -\infty < \mu < \infty \text{ and } 0 < \sigma < \infty\}$.

Normal Distribution – Estimate Likelihood Function

The joint probability function of $X_1, \dots, X_n$ is,

$$L(x_1, \dots, x_n; \mu, \sigma) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x_i - \mu)^2}{2\sigma^2}}$$

The likelihood function

$$L(\mu, \sigma) = \sigma^{-n} (2\pi)^{-n/2} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right]$$

OR

$$L(\mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sigma\sqrt{2\pi}} e^{-(x_i - \mu)^2 / (2\sigma^2)} = \frac{1}{(2\pi\sigma^2)^{n/2}} e^{-(1/2\sigma^2) \sum_{i=1}^n (x_i - \mu)^2}$$

Normal Distribution – Estimate Likelihood Function

Taking logarithm of the likelihood,

$$\ln L(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \log e$$

$$\log e = 1$$

$$\ln L(\mu, \sigma^2) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

Now taking derivative with respect to μ and set to 0,

$$\frac{d}{d\mu} \ln L(x_1, \dots, x_n; \mu, \sigma) = 0$$

$$\frac{d}{d\mu} \left(-\frac{n}{2} \ln(2\pi) - n \ln \sigma - \frac{(x_1 - \mu)^2}{2\sigma^2} - \dots - \frac{(x_n - \mu)^2}{2\sigma^2} \right)$$

$$= 0 - 0 + \frac{2(x_1 - \mu)}{2\sigma^2} + \dots + \frac{2(x_n - \mu)}{2\sigma^2} \quad \text{By applying chain rule,}$$

$$= \frac{1}{\sigma^2} (x_1 - \mu + \dots + x_n - \mu)$$

$$= \frac{1}{\sigma^2} [(x_1 + \dots + x_n) - n\mu]$$

Normal Distribution – Estimate of Mean

$$0 = \frac{1}{\sigma^2} [(x_1 + \cdots + x_n) - n\mu]$$

Solving for μ ,

$$\mu = \frac{(x_1 + \cdots + x_n)}{n}$$

$$\mu = \sum_{i=1}^n \frac{x_i}{n} = \bar{X}$$

The MLE of μ is $\hat{\mu} = \bar{X}$

Normal Distribution – Estimation of Variance

Now taking derivative with respect to σ and set to 0,

$$\frac{d}{d\sigma} \ln L(x_1, \dots, x_n; \mu, \sigma) = 0$$

$$\frac{d}{d\sigma} \left(-\frac{n}{2} \ln(2\pi) - n \ln \sigma - \frac{(x_1 - \mu)^2}{2\sigma^2} - \dots - \frac{(x_n - \mu)^2}{2\sigma^2} \right)$$

$$= 0 - \frac{n}{\sigma} + \frac{(x_1 - \mu)^2}{\sigma^3} + \dots + \frac{(x_n - \mu)^2}{\sigma^3} \quad \text{By applying chain rule,}$$

$$0 = -\frac{n}{\sigma} + \frac{1}{\sigma^3} [(x_1 - \mu)^2 + \dots + (x_n - \mu)^2]$$

Solving for σ ,

$$\sigma^2 = \frac{[(x_1 - \mu)^2 + \dots + (x_n - \mu)^2]}{n}$$

Normal Distribution – Estimation of Variance

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \mu)^2}{n}$$

Take square root on both sides ,

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (x_i - \mu)^2}{n}}$$

After substituting $\mu = \bar{X}$, we obtain

$$\sigma = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n}}$$

The MLE of σ is $\hat{\sigma} = \sigma = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n}}$

Normal Distribution – Estimate Likelihood Function

Problem:

Suppose you are given the following data from a sample assumed to be i.i.d. from a normal distribution:

$x = \{9, 11, 13, 12, 10\}$

Estimate the **MLE of mean (μ)** and **variance (σ^2)**.

Step 1: Compute sample mean

$$\bar{x} = \frac{9 + 11 + 13 + 12 + 10}{5} = \frac{55}{5} = \boxed{11}$$

Step 2: Compute MLE of variance

$$\begin{aligned}\hat{\sigma}^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \\ &= \frac{1}{5} [(9 - 11)^2 + (11 - 11)^2 + (13 - 11)^2 + (12 - 11)^2 + (10 - 11)^2] \\ &= \frac{1}{5} [4 + 0 + 4 + 1 + 1] = \frac{10}{5} = \boxed{2}\end{aligned}$$

Example – Maximum Likelihood Estimate for Mean (μ)

Problem:

Suppose the weights of randomly selected female college students are normally distributed with unknown mean μ and standard deviation σ . A random sample of 10 female college students yielded the following weights (in pounds):

115 122 130 127 149 160 152 138 149 180

Identify the likelihood function and the maximum likelihood estimator of μ , the mean weight of all female college students. Using the given sample, find a maximum likelihood estimate of μ as well.

Solution:

We know that, the MLE of μ is $\hat{\mu} = \bar{X}$

$$\hat{\mu} = \sum_{i=1}^n \frac{x_i}{n} = \bar{X}$$

$$\begin{aligned}\hat{\mu} &= \frac{1}{10} (115 + 122 + 130 + 127 + 149 + 160 + 152 + 138 + 149 + 180) \\ &= 142.2\end{aligned}$$

Given $x=\{1.2,0.9,1.5,1.1\}$ assume the data come from $N(\mu,\sigma^2)$. Compute the log-likelihood at the MLE estimates $\mu^\wedge,\sigma^{\wedge^2}$.

Solution

Step 1: Compute $\hat{\mu}$

$$\hat{\mu} = \frac{1.2 + 0.9 + 1.5 + 1.1}{4} = \boxed{1.175}$$

Step 2: Compute $\hat{\sigma}^2$

$$\hat{\sigma}^2 = \frac{1}{4}[(1.2 - 1.175)^2 + (0.9 - 1.175)^2 + (1.5 - 1.175)^2 + (1.1 - 1.175)^2] = \boxed{0.0525}$$

Step 3: Compute log-likelihood

$$\ln L = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\hat{\sigma}^2) - \frac{1}{2\hat{\sigma}^2} \sum (x_i - \hat{\mu})^2$$

Using $n = 4$, $\hat{\sigma}^2 = 0.0525$, $\sum (x_i - \hat{\mu})^2 = 0.21$:

$$\ln L = -2 \ln(2\pi) - 2 \ln(0.0525) - \frac{0.21}{2(0.0525)} \approx -2.756 - 5.948 - 2 = \boxed{-10.704}$$

Maximum likelihood is the most commonly used method of estimation.

The main reason for this is that in most cases that arise in practice, MLEs have two very desirable properties,

1. In most cases, as the sample size n increases, the bias of the MLE converges to 0.
2. In most cases, as the sample size n increases, the variance of the MLE converges to a theoretical minimum.

Pitfalls of Point Estimators



- A point estimator is a single number which may vary from sample to sample.
- Certainly the point estimators are slightly different from true population parameter .
- It cannot be confidently claimed to be close to the actual parameter.
- This can be solved by estimating population parameters in the given intervals of values where point estimator can be centered. This interval is called **confidence interval**.



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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 2: Confidence Intervals

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❖ Confidence Intervals - Introduction

Assume that a large number of independent measurements, using the same procedure, are made on the diameter of a piston.

The **sample mean** of the measurements is **14.0 cm**

The **standard deviation** of the sample mean is **0.1 cm**.

The value 14.0 comes from a **normal distribution**, because it is the average of a large number of measurements.

Now the **true diameter** of the piston **will certainly not be exactly equal to the sample mean of 14.0 cm**.

We can use its standard deviation to determine how close it is likely to be to the true diameter.

For example, it is very **unlikely** that the sample mean will differ from the true diameter by more than three standard deviations. Therefore we have a **high level of confidence** that the true diameter is in the interval **(13.7, 14.3)**.

On the other hand, it is **likely** for the sample mean to differ from the true value by more than one standard deviation. Therefore we have a **lower level of confidence** that the true diameter is in the interval **(13.9, 14.1)**.

The intervals (13.7, 14.3) and (13.9, 14.1) are **confidence intervals** for the true diameter of the piston.

- A 2008 Gallup survey found that TV ownership may be good for wellbeing. The results from the poll stated that the confidence level was 95% ± 3 , which means that if Gallup repeated the poll over and over, using the same techniques, 95% of the time the results would fall within the published results. The 95% is the confidence level and the ± 3 is called a margin of error.

- The U.S. Census Bureau routinely uses confidence levels of 90% in their surveys. One survey of the number of people in poverty in 1995 stated a confidence level of 90% for the statistics “The number of people in poverty in the United States is 35,534,124 to 37,315,094.” That means if the Census Bureau repeated the survey using the same techniques, 90 percent of the time the results would fall between 35,534,124 and 37,315,094 people in poverty. The stated figure (35,534,124 to 37,315,094) is the confidence interval.

- **Example:** A recent article on Rasmussen Reports states that “38% of Likely U.S. Voters now say their health insurance coverage has changed because of Obama care”.
- “The margin of sampling error is +/- 3 percentage points with a **95% level of confidence.**”
- What a 95% confidence level is saying is that **if the poll or survey were repeated over and over again, the results would match the results from the actual population 95% of the time.**

- The **width of the confidence interval** tells us more about **how certain (or uncertain) we are about the true figure in the population.**
- This width is stated as a plus or minus (in this case, +/- 3).
- In this case, you would expect the actual results to be between (35 , 41) or (38-3 , 38+3) percent, 95% of the time.

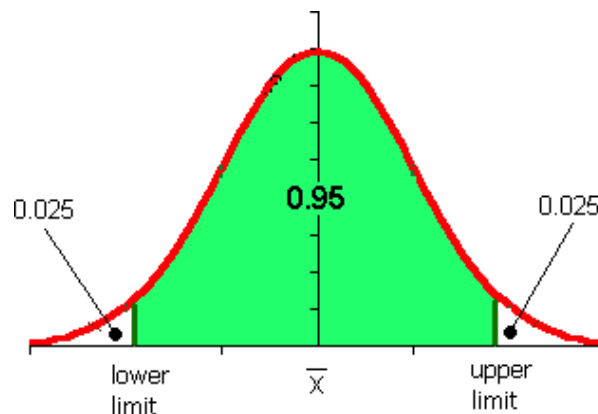
- **Population size**: Affects the CI when you are working with small groups.
- **Sample Size**: the smaller your sample, the less likely it is you can be confident the results reflect the true population parameter.
- **Percentage**: Extreme answers come with better accuracy.
- *For example*, if 99% of voters are for a particular party, the chances of error are small.
- However, if 49.9% of voters are “for” and 50.1 percent are “against” then the chances of error are bigger.

- A **0% confidence level** means you have **no faith at all** that if you repeated the survey that you would get the same results.
- A **100% confidence level** means there is **no doubt at all** that if you repeated the survey you would get the same results
- In reality, you would never publish the results from a survey where you had no confidence at all that your statistics were accurate.
- A 100% confidence level doesn't exist in statistics, unless you surveyed an entire population 100 percent sure that your survey wasn't open to some kind of error or bias.

- **A confidence interval is how much uncertainty there is with any particular statistic.**
- Confidence intervals are often used with a margin of error.
- It tells you how confident you can be that the results from a poll or survey reflect what you would expect to find if it were possible to survey the entire population.
- Confidence intervals are intrinsically connected to confidence levels.

- Confidence levels are expressed as a percentage (for example, a 95% confidence level).
- It means that should you repeat an experiment or survey over and over again, 95 percent of the time your results will match the results you get from a population (in other words, your statistics would be sound!).
- Confidence intervals are your results, usually the range in which you expect to find the results in .

- For example, you survey a group of pet owners to see how many cans of dog food they purchase a year.
- You test your statistics at the 99 percent confidence level and get a confidence interval of (200,300). That means you think they buy between 200 and 300 cans a year. You're super confident (99% is a very high level!) that your results are sound, statistically.



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Confidence Coefficient

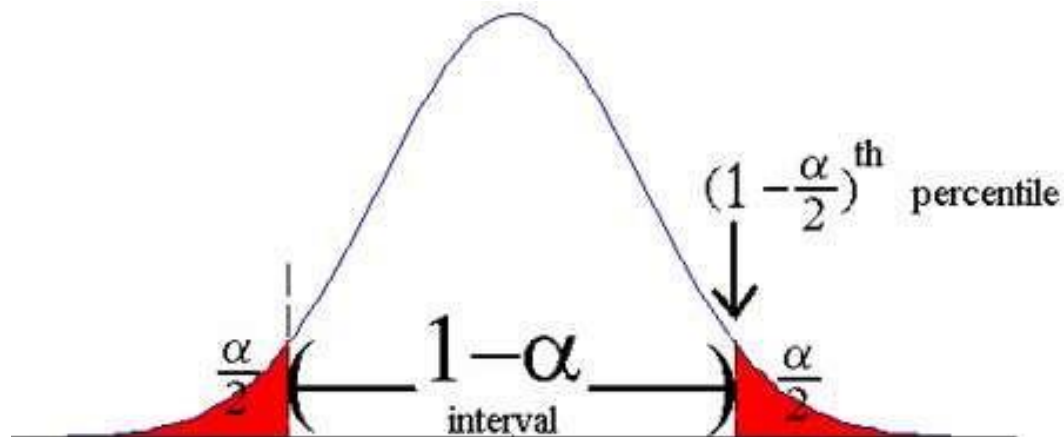
- The confidence coefficient is the confidence level stated as a proportion, rather than as a percentage.
- For example, if **confidence level = 99%**, **confidence coefficient = 0.99**.
- In general, the **higher the coefficient**, the **more certain you are that your results are accurate**.
- For example, a .99 coefficient is more accurate than a coefficient of .89.
- It's extremely rare to see a coefficient of 1 (meaning that you are positive without a doubt that your results are completely, 100% accurate).
- A coefficient of zero means that you have no faith that your results are accurate at all.

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Significance level

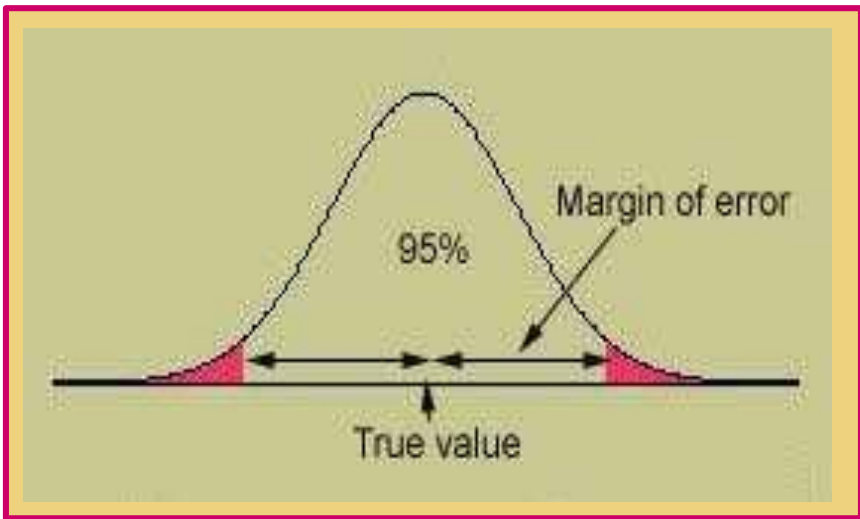
- Alpha(α) is called as the significance level.
- Confidence level is taken as $(1-\alpha)*100\%$

Confidence coefficient ($1 - \alpha$)	Confidence level ($1 - \alpha * 100\%$)	
0.90	90%	$\alpha = ?$
0.95	95%	$\alpha = ?$
0.99	99%	$\alpha = ?$



- **Margin of error: provides a bound to the difference between a particular estimate and the parameter that it estimates.**
- From the Central Limit Theorem, the sampling distributions of $\bar{x}$ and $\hat{p}$ will be approximately normal under certain assumptions
- For unbiased estimators with normal sampling distributions, **95% of all point estimates will lie within 1.96 standard deviations of the parameter of interest.**

Note:
Here
**we assume
that the
sample sizes
are large.**



So here,
Margin of error = $1.96 \times \text{std error of estimator}$

Why 1.96 ?

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Probability VS. Confidence



- A 95% confidence interval for the population mean μ was computed to be (12.304, 12.696).
- It is tempting to say that the probability is 95% that μ is between 12.304 and 12.696.
- **The term probability refers to random events, which can come out differently when experiments are repeated.**
- The numbers 12.304 and 12.696 are fixed, not random.
- The population mean is also fixed. The mean is either in the interval or it is not.
- There is no randomness involved. Therefore we say that we have 95% confidence (not probability) that μ is in this interval

Example

- A 90% confidence interval for the mean diameter (in cm) of steel rods manufactured on a certain extrusion machine is computed to be (14.73, 14.91). True or false: The probability that the mean diameter of rods manufactured by this process is between 14.73 and 14.91 is 90%.

- Solution
- False. A specific confidence interval is given. The mean is either in the interval or it isn't. We are 90% confident that the population mean is between 14.73 and 14.91.
- The term probability is inappropriate.

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Probability VS. Confidence



Let's say that we are discussing
a *method* used to compute a 95% confidence interval.

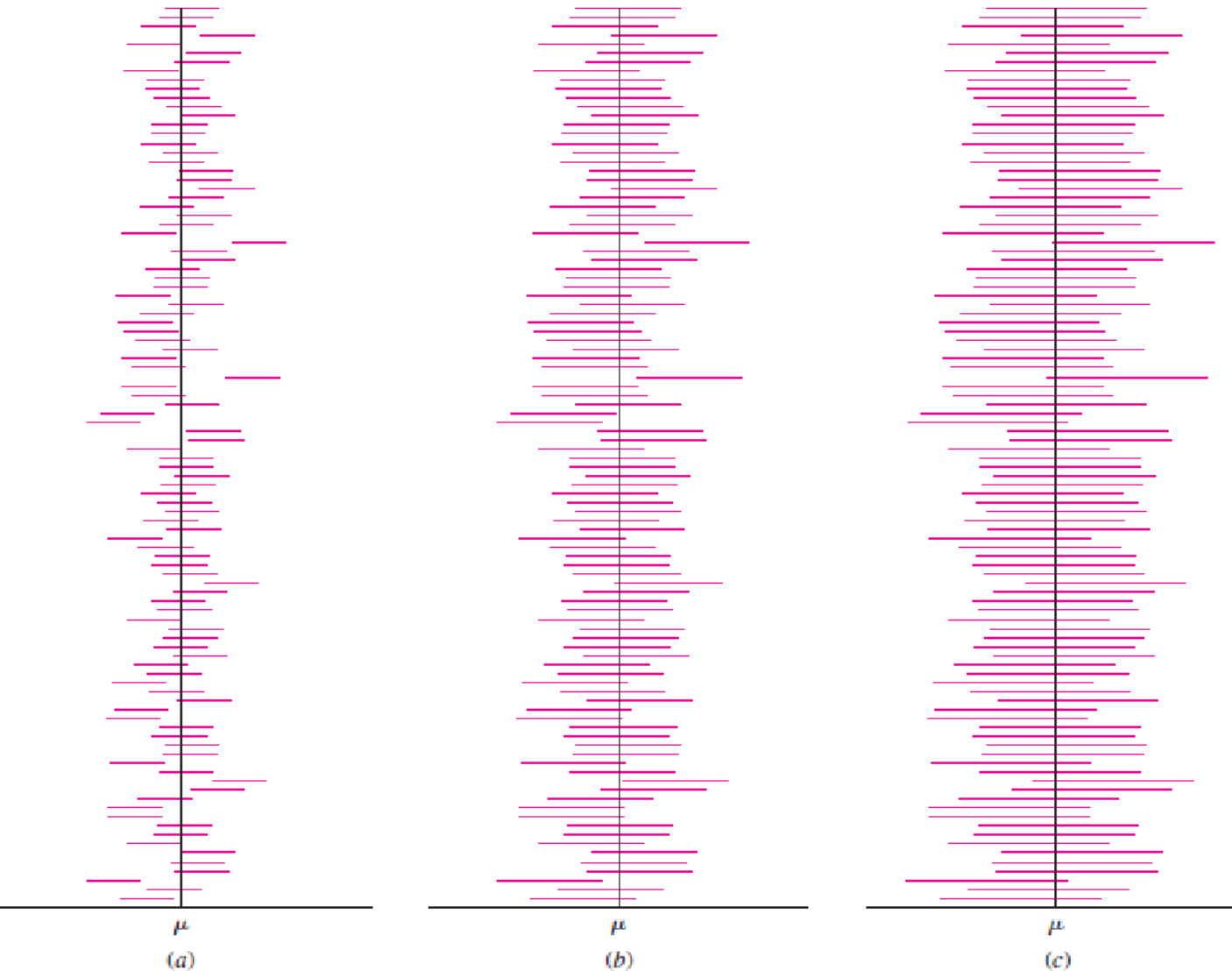
The method will succeed in covering the population mean 95% of the time,
and fail the other 5% of the time.

In this case, **whether the population mean is covered or not is a random event,**
because it can vary from experiment to experiment.

Therefore it *is* **correct** to say that
**a *method* for computing a 95% confidence interval has probability 95% of
covering the population mean.**

- ❖ Probability and confidence, while related, represent different concepts in statistics.
- ❖ Probability refers to the likelihood of a specific event occurring, while confidence reflects the degree of certainty that a statistical estimate (like a confidence interval) contains the true population parameter.
- ❖ A 95% confidence interval means that if you were to repeat the sampling process many times, 95% of the calculated confidence intervals would contain the true population parameter.

Confidence focuses on the reliability of an interval estimate, not on the probability of a single event.



(a) One hundred 68% confidence intervals for a population mean, each computed from a different sample. Although precise, they fail to cover the population mean 32% of the time. This high failure rate makes the 68% confidence interval unacceptable for practical purposes.

(b) One hundred 95% confidence intervals computed from these samples. This represents a good compromise between reliability and precision for many purposes.

(c) One hundred 99.7% confidence intervals computed from these samples. These intervals fail to cover the population mean only three times in 1000. They are extremely reliable, but imprecise.

An engineer plans to compute a 90% confidence interval for the mean diameter of steel rods. She will measure the diameters of a large sample of rods, compute $\bar{X}$ and s , and then compute the interval $\bar{X} \pm 1.645s/\sqrt{n}$.

True or false:

The probability that the population mean diameter will be in this interval is 90%.

Solution

- **True.** What is described here is a method for computing a confidence interval, rather than a specific numerical value. It is correct to say that a method for computing a 90% confidence interval has probability 90% of covering the population mean.

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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 2: Confidence Intervals

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Confidence Intervals: Introduction

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❖ One sided and Two-Sided Confidence Intervals

Two-Sided and one-sided Confidence Intervals

- ❖ Confidence intervals estimate a range of plausible values for an unknown population parameter.
- ❖ Two-sided intervals capture the parameter from both lower and upper bound, while one-sided intervals provide either a lower or upper bound.
- ❖ The choice between one-sided and two-sided intervals depends on the research question and the directionality of the alternative hypothesis.

Two-Sided and one-sided Confidence Intervals

- ❖ The confidence interval involves capturing the potential range around the sample mean. The formula for a two-sided confidence interval is:
- ❖ The formula for a two-sided confidence interval is generally:

Sample Statistic $\pm$ (Critical Value * Standard Error).

Two-Sided and one-sided Confidence Intervals

For one-sided hypothesis tests, where the researcher is interested in testing if a parameter is either greater or lesser than a certain value, the confidence interval can be tailored to reflect this focus. To construct a confidence interval for the lower or upper limit, we use the following formulas:

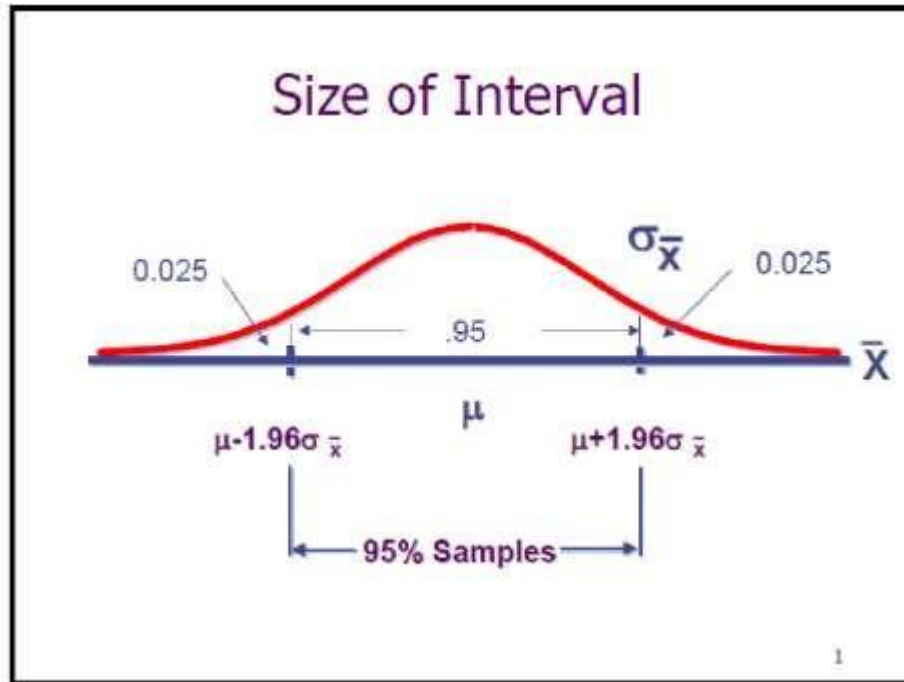
- **For a lower one-sided interval:**

Lower Limit = Sample Mean – (Critical Value × Standard Error)

- **For an upper one-sided interval:**

Upper Limit = Sample Mean + (Critical Value × Standard Error)

Two-Sided Confidence Intervals



Two-Sided C. I.

$$Z \text{ C.I.: } \left(\bar{X} - Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}, \bar{X} + Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} \right)$$

One-Sided Confidence Intervals

Z C.I.:

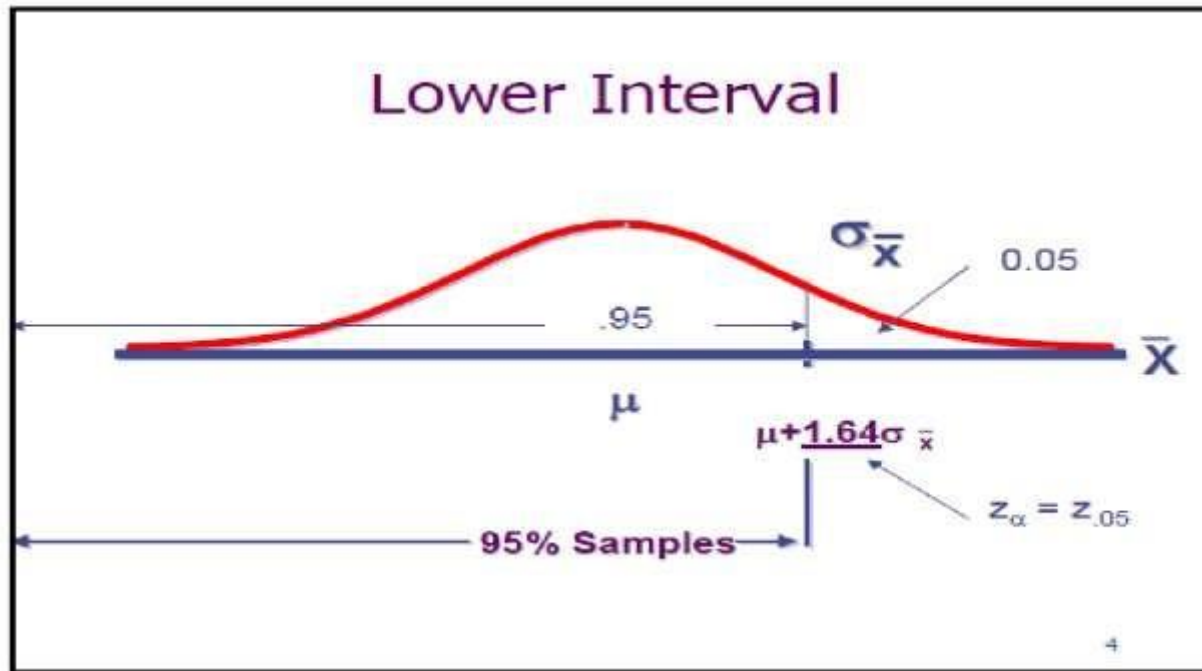
Lower interval $(-\infty, \bar{X} + Z_{\alpha} \cdot \frac{s}{\sqrt{n}})$

Upper interval $(\bar{X} - Z_{\alpha} \cdot \frac{s}{\sqrt{n}}, \infty)$

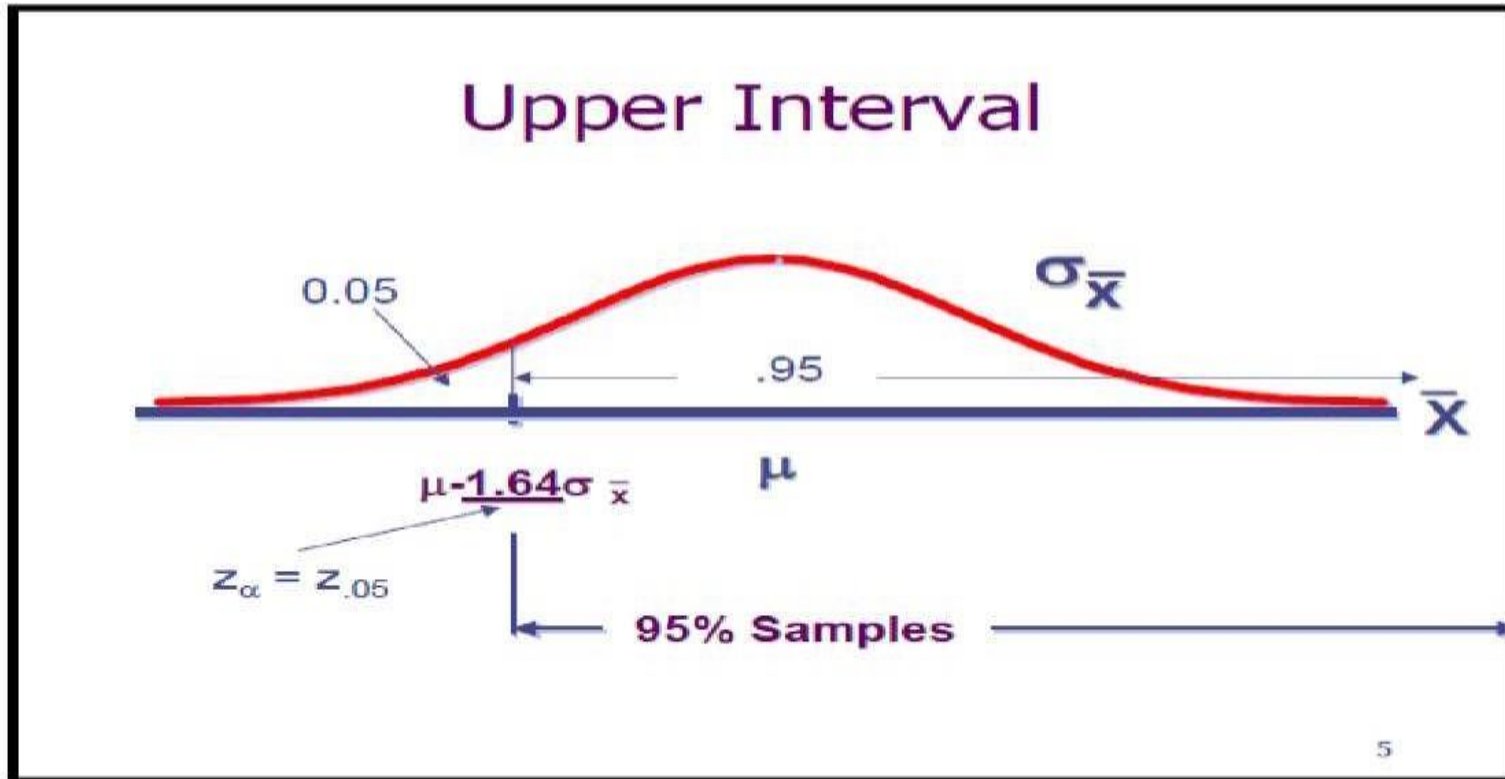
Lower bound

Upper bound

One-Sided Confidence Intervals



One-Sided Confidence Intervals



Two-Sided Confidence Intervals

❖ If a researcher wants to estimate the average height of adult males, they would use a two-sided confidence interval to determine the range of plausible heights.

Solution:

A two-sided confidence interval uses critical values from both tails of the distribution. For example, a 95% two-sided interval uses critical values from the 2.5th and 97.5th percentiles.

One-Sided Confidence Intervals

❖ If a company wants to ensure their product's average lifespan is at least 1000 hours, they would use a one-sided confidence interval to determine if there's sufficient evidence to support this claim.

Solution:

A one-sided confidence interval uses a critical value corresponding to a single tail of the distribution.

For example, a 95% one-sided interval uses a critical value from the 95th percentile for an upper bound or the 5th percentile for a lower bound.

One-Sided Confidence Intervals

- ❖ A researcher wants to estimate the average height of adult males in a city. They want to know both a lower and upper bound for the average height.

Solution: A 95% two-sided confidence interval is calculated

- ❖ A pharmaceutical company is testing a new drug and wants to be 95% confident that the drug's effectiveness is at least a certain level. They only care about the lower bound of effectiveness.

Solution: A 95% one-sided confidence interval is calculated

One-Sided Confidence Intervals

❖ Let's say a study on the average income of residents in a town found a sample mean of \$50,000 with a standard error of \$2,000. Then,

❖ **95% Two-sided CI:**

$\$50,000 \pm (1.96 * \$2,000) = \$50,000 \pm \$3,920$. The interval would be (\$46,080, \$53,920).

❖ **95% One-sided CI (lower bound):**

$\$50,000 - (1.645 * \$2,000) = \$50,000 - \$3,290 = \$46,710$.
The interval would be (\$46,710, ∞).

One-Sided Confidence Intervals

- ❖ For various confidence levels, the critical values for two-sided and one-sided hypothesis tests are different.
- ❖ For a two-sided test, the critical values are found by splitting the alpha level ($1 - \text{confidence level}$) across both tails of the distribution.
- ❖ For a one-sided test, the entire alpha level is placed in either the left or right tail

Two-Sided Confidence Intervals

Critical Values (z-values) for Confidence Intervals (Two-Sided)

Confidence Level	Critical Value (z)
90%	1.645
95%	1.96
98%	2.33
99%	2.575

One-Sided Confidence Intervals

Critical Values (z-values) for One-Sided Hypothesis Tests

Confidence Level	Critical Value (z)
90%	1.28
95%	1.645
98%	2.05
99%	2.33

One-Sided Confidence Intervals

Summary

Let $X_1, \dots, X_n$ be a *large* ($n > 30$) random sample from a population with mean μ and standard deviation σ , so that $\bar{X}$ is approximately normal. Then level $100(1 - \alpha)\%$ lower confidence bound for μ is

$$\bar{X} - z_{\alpha}\sigma_{\bar{X}} \quad (5.2)$$

and level $100(1 - \alpha)\%$ upper confidence bound for μ is

$$\bar{X} + z_{\alpha}\sigma_{\bar{X}} \quad (5.3)$$

where $\sigma_{\bar{X}} = \sigma/\sqrt{n}$. When the value of σ is unknown, it can be replaced with the sample standard deviation s .

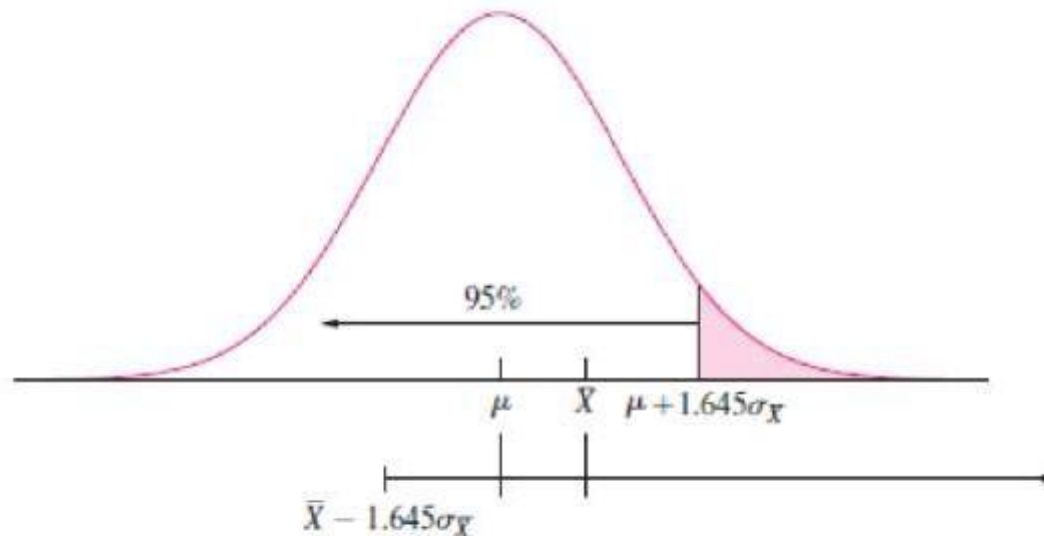


FIGURE 5.6 The sample mean $\bar{X}$ is drawn from a normal distribution with mean μ and standard deviation $\sigma_{\bar{X}} = \sigma/\sqrt{n}$. For this particular sample, $\bar{X}$ comes from the lower 95% of the distribution, so the 95% one-sided confidence interval $(\bar{X} - 1.645\sigma_{\bar{X}}, \infty)$ succeeds in covering the population mean μ .

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References



- “Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 4th Edition, 2015.



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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 2: Confidence Intervals

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❖ 5.1 - Large-Sample Confidence Intervals for a Population Mean

- ❖ Confidence intervals for large samples provide a range of plausible values for a population parameter, such as the mean or proportion, based on a sample from that population.
- ❖ For large samples (typically $n \geq 30$), these intervals are often constructed using the normal distribution, due to the Central Limit Theorem.
- ❖ The confidence level indicates the probability that the interval will contain the true population parameter if the sampling process were repeated many times.

Confidence Interval for the Mean

Point Estimate $\pm$ Margin of Error

Point Estimate $\pm$ (Critical Value)(Standard Error)

$$\bar{x} \pm (\text{Critical Value})(\sigma_{\bar{X}})$$

- The point estimate of the parameter of interest is the calculated statistic, in this case, $\bar{x}$.
- The critical value is determined from the confidence level.
- The standard error is the standard deviation of the statistic, in this case,

$$\sigma_{\bar{X}}$$

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Confidence Intervals for large samples

- If σ is known:

$$\bar{x} \pm z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right)$$

- If σ is unknown:

$$\bar{x} \pm z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$$

A sample is considered large when $n \geq 30$.

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Confidence Intervals for large samples

Table 1: Critical Values from the Standard Normal Distribution

Confidence Level	α	$\alpha/2$	$z_{\alpha/2}$	Confidence Interval Formula
68%	0.32	0.16	1.00	$\bar{x} \pm 1\sigma_{\bar{x}}$
90%	0.10	0.05	1.645	$\bar{x} \pm 1.645\sigma_{\bar{x}}$
95%	0.05	0.025	1.96	$\bar{x} \pm 1.96\sigma_{\bar{x}}$
99%	0.01	0.005	2.576	$\bar{x} \pm 2.576\sigma_{\bar{x}}$
99.7%	0.003	0.0015	2.97	$\bar{x} \pm 2.97\sigma_{\bar{x}}$

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Estimating Means and Proportions

- **For a Quantitative population,**

Note:

Large Sample: $n \geq 30$

Point estimator of population mean μ : $\{\bar{x}$

Margin of error ($n \geq 30$): $\pm 1.96 \frac{s}{\sqrt{n}}$ (given that confidence level = 95%)

- **For a Binomial population,**

Point estimator of population proportion p : $\{\hat{p} = x/n$

Margin of error: $\pm 1.96 \sqrt{\frac{\hat{p}\hat{q}}{n}}$ Assumption: $np > 5$ and $nq > 5$; or $0 < p \pm 2\sqrt{\frac{pq}{n}} < 1$

(given that confidence level = 95%)

A homeowner randomly **samples 64** homes similar to her own and finds that the **average selling price is \$250,000** with a **standard deviation of \$15,000**. Estimate the average selling price for all similar homes in the city.

A homeowner randomly samples 64 homes similar to her own and finds that the average selling price is \$250,000 with a standard deviation of \$15,000. Estimate the average selling price for all similar homes in the city.

Point estimator of μ : $\{\bar{x} = 250,000$

$$\text{Margin of error: } \pm 1.96 \frac{s}{\sqrt{n}} = \pm 1.96 \frac{15,000}{\sqrt{64}} = \pm 3675$$

$$\text{Confidence Interval} = \bar{x} \pm \text{MoE} = 250,000 \pm 3,675$$

$$\text{Confidence Interval} = (246,325, 253,675)$$

- Create an interval (a, b) so that you are fairly sure that the parameter lies between these two values.
- “Fairly sure” means “with high probability”, measured using the confidence coefficient, $(1 - \alpha)$

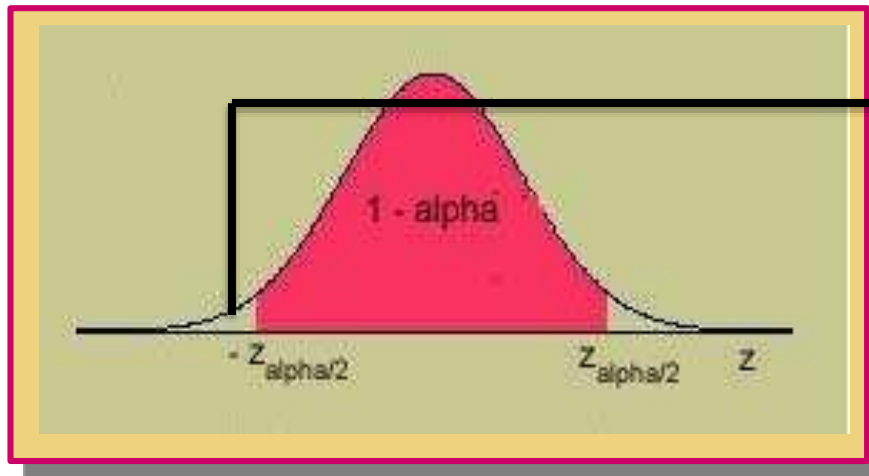
Usually, $1 - \alpha = .90, .95, .99$

- For large-Sample size,

100(1- α)% Confidence Interval:

Point Estimator $\pm z_{\alpha/2}SE$

- To change to a general confidence level, $(1-\alpha)*100\%$, pick a value of z that puts area $1-\alpha$ (confidence coeff.) in the center of the z distribution.



Tail area $\alpha/2$	$z_{\alpha/2}$
.05	1.645
.025	1.96
.005	2.58

Table entries for z represent the area under the bell curve to the left of z

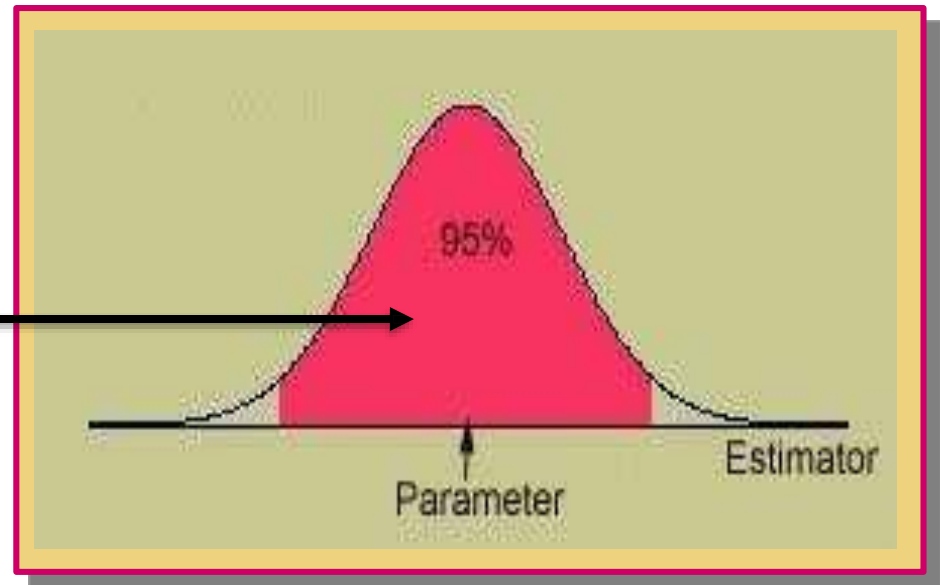
Ex: for $\alpha = 0.5$ (confidence level = 95%), $\alpha/2 = 0.025$, find this area in the Z table. The area 0.025 corresponds to Z value of 1.96.

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To Change the Confidence Level

- Suppose $1-\alpha = .95$,

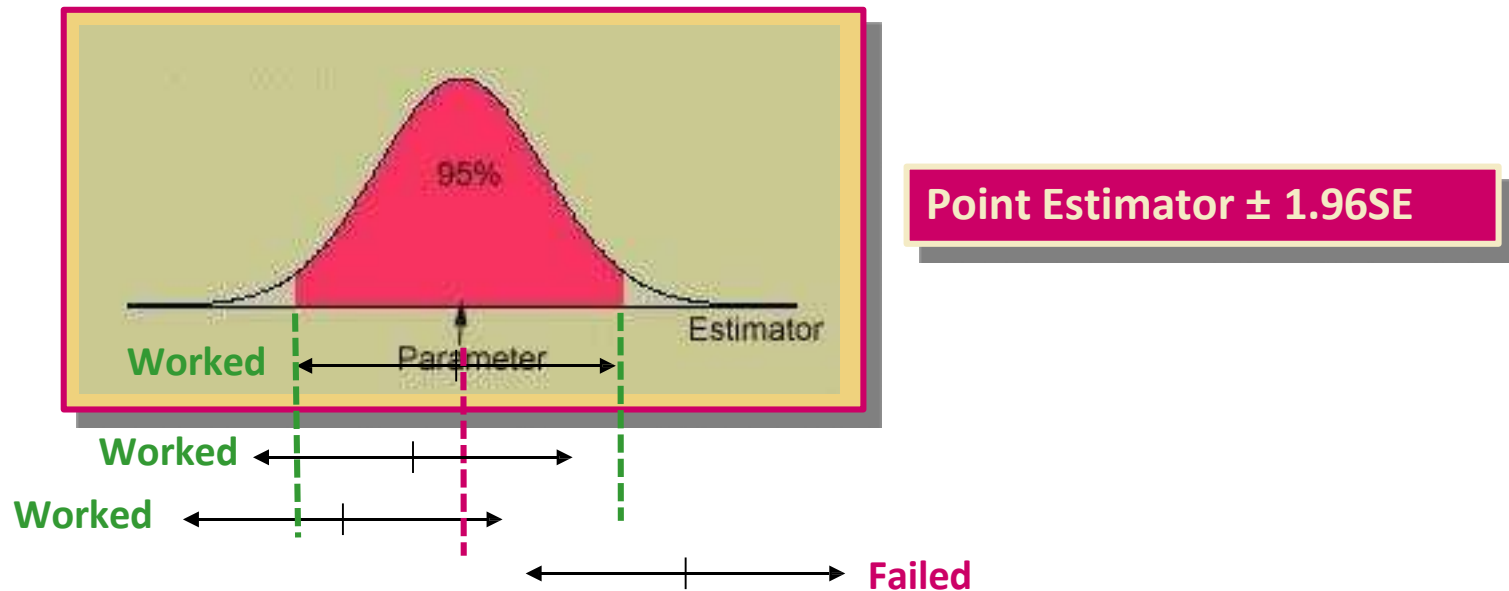
95% of the intervals constructed in this manner will enclose the population mean



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To Change the Confidence Level

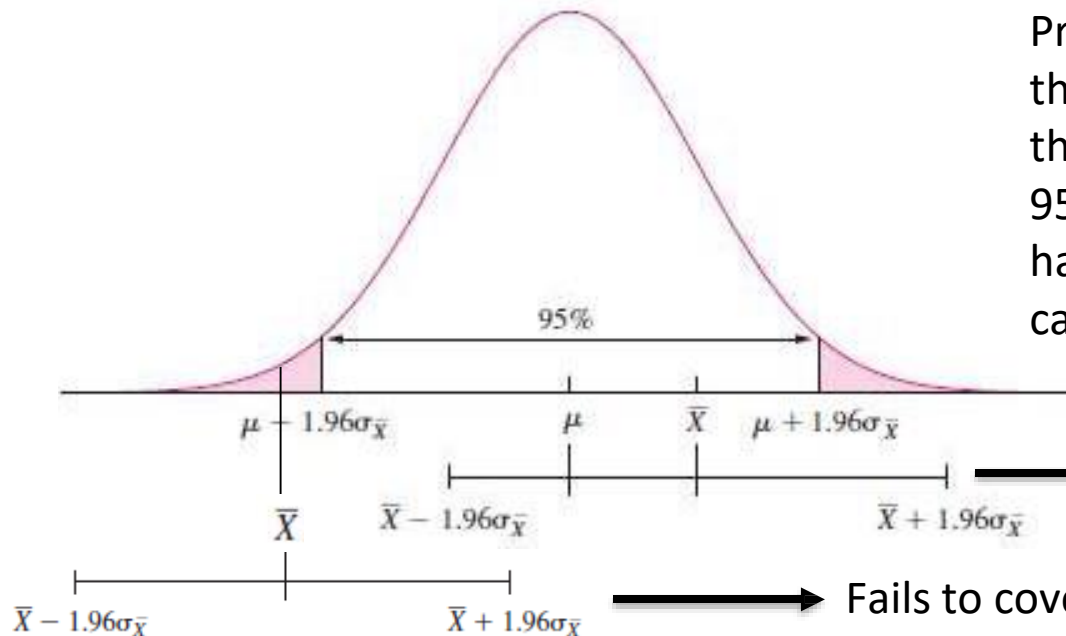
- Since we don't know the value of the parameter, consider which has a variable center.



- Only if the estimator falls in the tail areas will the interval fail to enclose the parameter. This happens only 5% of the time.

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Interval Estimation/Confidence Interval



Presents one possibility, which is that the sample mean $\bar{X}$ lies within the middle 95% of the distribution. 95% of all the samples that could have been drawn fall into this category.

Sample succeeds in covering μ

Fails to cover μ

Represents a sample whose mean $\bar{X}$ lies outside the middle 95% of the curve. Only 5% of all the samples that could have been drawn fall into this category. For these more unusual samples the 95% confidence interval $\bar{X} \pm 1.96\sigma_{\bar{X}}$ fails to cover the population mean μ .

- A confidence interval is calculated from one given sample. It either covers or misses the true parameter. Since the true parameter is unknown, you'll never know which one is true.
- If independent samples are taken repeatedly from the same population, and a confidence interval calculated for each sample, then a certain percentage (confidence level) of the intervals will include the unknown population parameter.
- The confidence level associated with a confidence interval is the success rate of the confidence interval.

A random sample of $n = 50$ males showed a **mean average** daily intake of dairy products equal to **756 grams** with a **standard deviation of 35 grams**. Find a

- i) 95% confidence interval
 - ii) 99% confidence interval
- for the population average μ

A random sample of **n = 50** males showed a **mean average** daily intake of dairy products equal **to 756 grams** with a **standard deviation of 35 grams**.

Find a

- i) 95% confidence interval
 - ii) 99% confidence interval
- for the population average μ

$$\bar{x} \pm 1.96 \frac{s}{\sqrt{n}} \Rightarrow 756 \pm 1.96 \frac{35}{\sqrt{50}} \Rightarrow 756 \pm 9.70$$

$$\text{or } 746.30 < \mu < 765.70 \text{ grams.}$$

ii) 99% CI

$$\bar{x} \pm 2.58 \frac{s}{\sqrt{n}} \Rightarrow 756 \pm 2.58 \frac{35}{\sqrt{50}} \Rightarrow 756 \pm 12.77$$

or $743.23 < \mu < 768.77$ grams.

Why is the confidence interval larger for 99% than 95% ?

The interval must be wider to provide for the increased confidence, that is does indeed enclose the true value of μ .

A random sample of $n = 150$ college students, **104 of the students** said that they had played on a soccer team during their K-12 years. Estimate the proportion of college students who played soccer in their youth with a 90% confidence interval.

- Sample proportion:

$$\hat{p} = \frac{104}{150} \approx 0.693$$

- Standard error:

$$SE = \sqrt{\frac{0.693(1-0.693)}{150}} \approx 0.0376$$

- For 90% CI, $z = 1.645$.

Margin of error:

$$ME = 1.645 \times 0.0376 \approx 0.062$$

- Confidence interval:

$$0.693 \pm 0.062 = (0.631, 0.755)$$

- Do you think we desire narrow confidence intervals or wide confidence intervals?
(Ans: Narrow, Why?)
- Do you think we desire low confidence level or high confidence level?
(Ans: High, Why?)
- Do you think we desire narrow confidence intervals with high confidence level or wide confidence intervals with low confidence level?

Ans: We aim for **narrow confidence intervals with high confidence levels** because they provide both precision and reliability in our estimates. However, achieving this balance often requires collecting more data or improving the precision of the sample.

- We are 95% confident that the average GPA of all college students is between 1.0 and 4.0.
- We are 95% confident that the average GPA of all college students is between 2.7 and 2.9.
- Which of these two intervals is more informative?
- In general, the narrower the confidence interval, the more information we have about the value of the population parameter.
- Therefore, we want all of our confidence intervals to be as narrow as possible.

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Width of a Confidence Interval

- To find the width of the confidence interval, we just take the difference in the two limits:

$$\text{Width} = \text{Upper Limit} - \text{Lower Limit} = 2 \times Z_{\alpha/2} (s/\sqrt{n})$$

If $\bar{X}$ increases, width is unaffected.

If s decreases, width decreases.

If confidence level decreases, width decreases Why?

If n increases, width decreases

- As the sample mean increases, the length stays the same. That is, the sample mean plays no role in the width of the interval.
- As the sample standard deviation s decreases, the width of the interval decreases. Since s is an estimate of how much the data vary naturally
- As we decrease the confidence level, the z -multiplier decreases, and hence the width of the interval decreases. In practice, we wouldn't want to set the confidence level below 90%.
- As we increase the sample size, This is the factor that we have the most flexibility in changing, the width of the interval decreases

A 95% confidence interval for a population mean is calculated from a sample size of 400. A second 95% confidence interval is calculated from a sample drawn from the same population with a sample size of 100. Compare the widths for the resulting confidence intervals.

A 95% confidence interval for the population mean is calculated as $\bar{x} \pm z_{0.025} \frac{\sigma}{\sqrt{n}}$. Notice we do not have any data to use so we will analyze these intervals symbolically. We do know $\alpha = 0.05$, so the $z_{\frac{\alpha}{2}}$ critical value is 1.96.

Sample Size of 400: $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{400}}$ gives $\bar{x} \pm 1.96 \frac{\sigma}{20}$.

Sample Size of 100: $\bar{x} \pm 1.96 \frac{\sigma}{\sqrt{100}}$ gives $\bar{x} \pm 1.96 \frac{\sigma}{10}$.

Notice how changing the sample size affects the width of the interval. A sample size of 100 produces an interval twice as wide. The interval produced from the sample size of 400 will be half as wide. The greater the sample size, the better the estimates of the population parameters, so the narrower the intervals.

A random sample of 50 adult females was taken and their RBC count is measured. The sample mean is 4.63 and the standard deviation of RBC count is 0.54. Construct a 95% confidence interval estimate for the true mean RBC count in adult females.

Solution:

For a 95% confidence interval the z-value is 1.960.

Apply z-value in the formula: $\mu \pm z \left(\frac{\sigma}{\sqrt{n}} \right)$

The confidence interval becomes: (4.480,4.780).

Therefore we are **95% confident** that the true mean RBC count for adult females is between **4.480** and **4.780**.

Summary

Let $X_1, \dots, X_n$ be a *large* ($n > 30$) random sample from a population with mean μ and standard deviation σ , so that $\bar{X}$ is approximately normal. Then a level $100(1 - \alpha)\%$ confidence interval for μ is

$$\bar{X} \pm z_{\alpha/2} \sigma_{\bar{X}} \quad (5.1)$$

where $\sigma_{\bar{X}} = \sigma / \sqrt{n}$. When the value of σ is unknown, it can be replaced with the sample standard deviation s .

In particular,

- $\bar{X} \pm \frac{s}{\sqrt{n}}$ is a 68% confidence interval for μ .
- $\bar{X} \pm 1.645 \frac{s}{\sqrt{n}}$ is a 90% confidence interval for μ .
- $\bar{X} \pm 1.96 \frac{s}{\sqrt{n}}$ is a 95% confidence interval for μ .
- $\bar{X} \pm 2.58 \frac{s}{\sqrt{n}}$ is a 99% confidence interval for μ .
- $\bar{X} \pm 3 \frac{s}{\sqrt{n}}$ is a 99.7% confidence interval for μ .

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References



- “Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 4th Edition, 2015.



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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 2: Confidence Intervals

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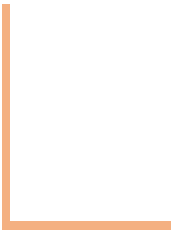
Confidence Intervals: One Sided

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- ❖ 5.1 - Large-Sample Confidence Intervals for a Population Mean- One Sided

One-Sided Confidence Intervals

A one-sided confidence interval estimates a population parameter by providing either a lower or upper bound, but not both. It is used when only one threshold (minimum or maximum) is of practical interest in decision-making.

A **one-sided confidence interval** focuses on just one direction:

- A **lower one-sided confidence interval** tells us the parameter is **greater than or equal to** a value.
- An **upper one-sided confidence interval** tells us the parameter is **less than or equal to** a value.

Need for One-Sided Confidence Intervals- Example

Let's say a **computer science engineer** is designing a **real-time communication system** (e.g., for online gaming or live video conferencing), where **low network latency** is critical.

The engineer collects a sample of latency data (in milliseconds) and wants to ensure that the **mean network latency** does **not exceed** a certain threshold — say, 100 ms — to meet real-time performance requirements.

Here, only the **upper bound** matters.

So, the engineer constructs an **upper one-sided 95% confidence interval** for the mean latency. If the interval shows latency ≤ 85 ms, they can confidently meet the 100 ms threshold.

One-Sided Confidence Intervals

Summary

Let $X_1, \dots, X_n$ be a *large* ($n > 30$) random sample from a population with mean μ and standard deviation σ , so that $\bar{X}$ is approximately normal. Then level $100(1 - \alpha)\%$ lower confidence bound for μ is

$$\bar{X} - z_{\alpha}\sigma_{\bar{X}} \quad (5.2)$$

and level $100(1 - \alpha)\%$ upper confidence bound for μ is

$$\bar{X} + z_{\alpha}\sigma_{\bar{X}} \quad (5.3)$$

where $\sigma_{\bar{X}} = \sigma/\sqrt{n}$. When the value of σ is unknown, it can be replaced with the sample standard deviation s .

One-Sided Confidence Intervals

Z C.I.:

Lower interval $(-\infty, \bar{X} + Z_{\alpha} \cdot \frac{s}{\sqrt{n}})$

Upper interval $(\bar{X} - Z_{\alpha} \cdot \frac{s}{\sqrt{n}}, \infty)$

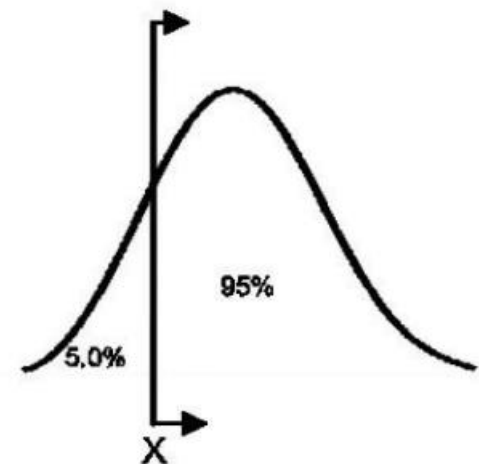
Lower bound

Upper bound

One-Sided Confidence Intervals

An **upper one-sided bound** defines a point that a certain percentage of the population is greater than.

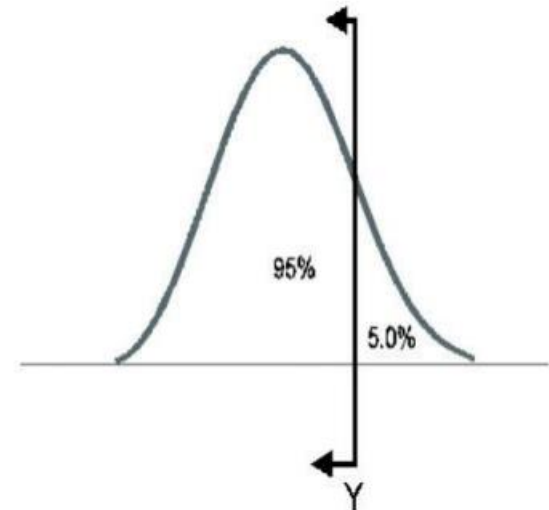
For example, for a 95% upper one-sided bound, this would indicate that 95% of the population parameter is greater than X .



One-Sided Confidence Intervals

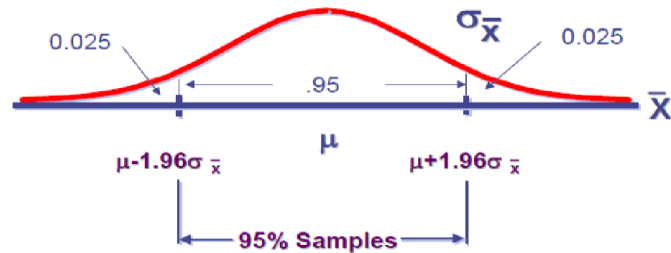
A **lower one-sided bound** defines a point that a specified percentage of the population is less than.

For example, For a 95% lower one-sided bound, this would indicate that 95% of the population is less than Y



Comparison of Two-Sided & One-Sided Confidence Intervals

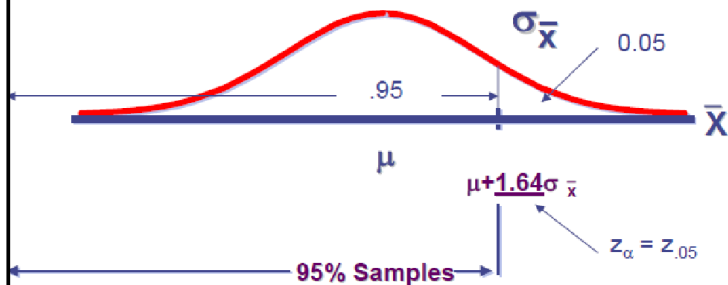
Size of Interval



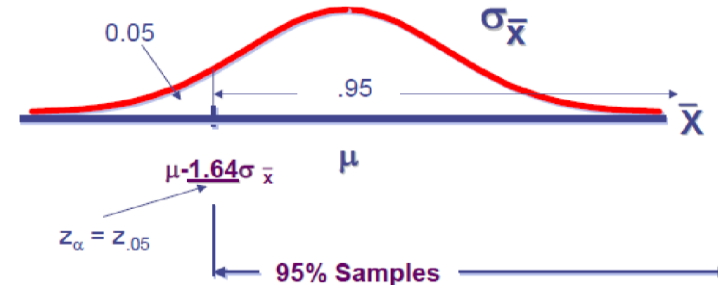
Two-Sided C. I.

$$Z \text{ C.I.: } \left(\bar{X} - Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}, \bar{X} + Z_{\alpha/2} \cdot \frac{s}{\sqrt{n}} \right)$$

Lower Interval



Upper Interval



Example

In a sample of 50 microdrills drilling a low-carbon alloy steel, the average lifetime (expressed as the number of holes drilled before failure) was 12.68 with a standard deviation of 6.83. **Find both a 95% lower confidence bound and a 99% upper confidence bound for the mean lifetime of the microdrills.**

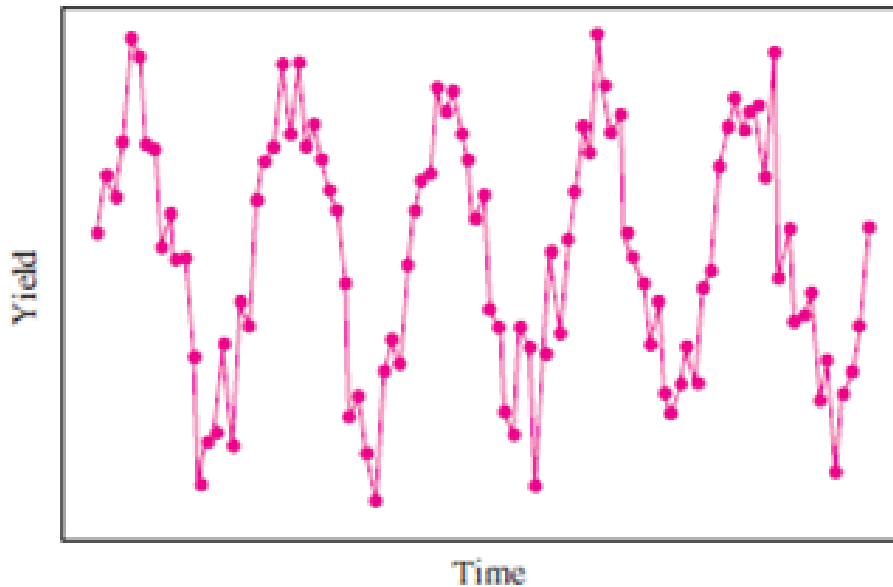
Solution

- The sample mean and standard deviation are $\bar{X} = 12.68$ and $s = 6.83$, respectively. The sample size is $n = 50$.
- We estimate $\sigma_{\bar{X}} \approx s/\sqrt{n} = 0.9659$.
- **The 95% lower confidence bound is $\bar{X} - 1.645\sigma_{\bar{X}} = 11.09$, and the 99% upper confidence bound is $\bar{X} + 2.33\sigma_{\bar{X}} = 14.93$.**

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Confidence Intervals Must Be Based on Random Samples

A chemical engineer wishes to estimate the mean yield of a new process. The process is run 100 times over a period of several days. Figure presents the 100 yields plotted against time. **Would it be appropriate to compute a confidence interval for the mean yield by calculating $\bar{X}$ and s for the yields.**



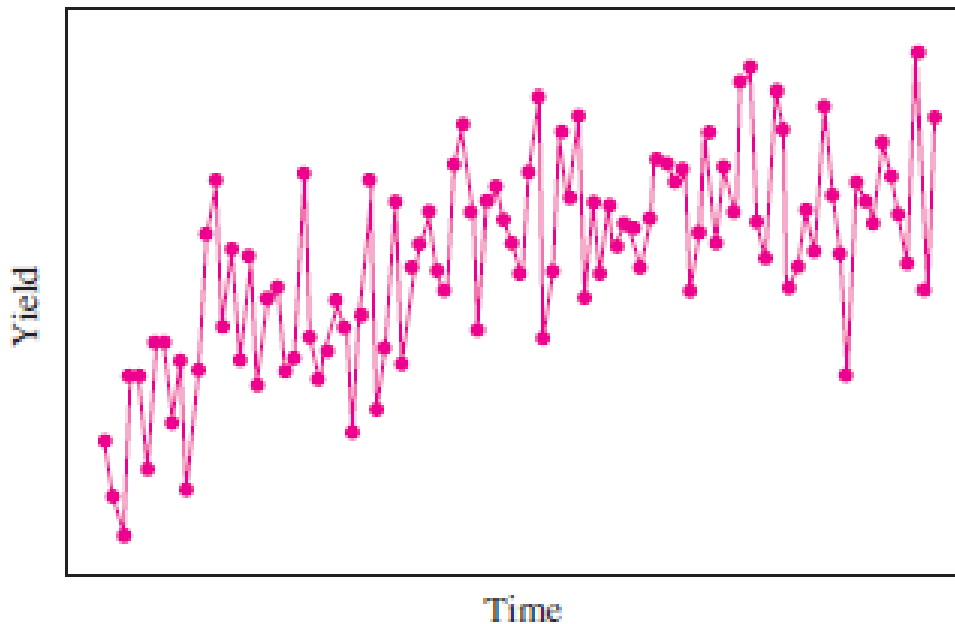
No. The formulae are valid only when the data are a random sample from a population. **Figure shows a cyclic pattern.**

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Confidence Intervals Must Be Based on Random Samples

The engineer in Example is studying the yield of another process.

Figure presents yields from 100 runs of this process, plotted against time. compute a confidence interval for the mean yield of this process?



No. The formulae are valid only when the data are a random sample from a population.
Figure shows a pattern.

A large sample of size $n=100$ is taken from a population with unknown mean and known standard deviation $\sigma=10$. The sample mean is $\bar{x}=52.3$. What is the 99% one-sided upper confidence bound for the population mean?

- A. $\mu < 52.3 + 2.33$
- B. $\mu > 52.3 - 2.58$
- C. $\mu < 52.3 + 2.58$
- D. $\mu > 52.3 - 1.96$

Solution: C

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References



“Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 6th Edition, 2024.



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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 2: Confidence Intervals

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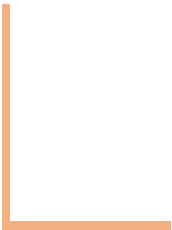
Confidence Intervals for Proportions

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- ❖ 5. 3- Confidence Intervals for Proportions (Traditional and Agresti Coull Method)

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Confidence Intervals for Proportions

Traditional method:

Summary

The Traditional Method for Computing Confidence Intervals for a Proportion (widely used but not recommended)

Let $\hat{p}$ be the proportion of successes in a *large* number n of independent Bernoulli trials with success probability p . Then the traditional level $100(1 - \alpha)\%$ confidence interval for p is

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \quad (5.8)$$

The method cannot be used unless the sample contains at least 10 successes and 10 failures.

Note:

The **traditional method cannot be used for small samples at all**; one rule of thumb regarding the sample size is **that both $n\hat{p}$ (the number of successes) and $n(1 - \hat{p})$ (the number of failures) should be greater than 10**.

- The **traditional method** of computing confidence interval **was justified on the basis of the Central Limit Theorem, which requires n to be large.**
- However, **the Agresti Coull method of computing confidence intervals is appropriate for any sample size n .**
- When used with small samples, it may occasionally happen that the lower limit is less than 0 or that the upper limit is greater than 1.
- Since $0 < p < 1$, a lower limit less than 0 should be replaced with 0, and an upper limit greater than 1 should be replaced with 1.

The confidence interval given by expression (5.5) is sometimes called the Agresti Coull interval, after Alan Agresti and Brent Coull, who developed it.

Summary

Let X be the number of successes in n independent Bernoulli trials with success probability p , so that $X \sim \text{Bin}(n, p)$.

Define $\tilde{n} = n + 4$, and $\tilde{p} = \frac{X + 2}{\tilde{n}}$. Then a level $100(1 - \alpha)\%$ confidence interval for p is

$$\tilde{p} \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{\tilde{n}}} \quad (5.5)$$

If the lower limit is less than 0, replace it with 0. If the upper limit is greater than 1, replace it with 1.

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Agresti Coull confidence interval

Summary

Let X be the number of successes in n independent Bernoulli trials with success probability p , so that $X \sim \text{Bin}(n, p)$.

Define $\tilde{n} = n + 4$, and $\tilde{p} = \frac{X + 2}{\tilde{n}}$. Then a level $100(1 - \alpha)\%$ lower confidence bound for p is

$$\tilde{p} - z_{\alpha} \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{\tilde{n}}} \quad (5.6)$$

and level $100(1 - \alpha)\%$ upper confidence bound for p is

$$\tilde{p} + z_{\alpha} \sqrt{\frac{\tilde{p}(1 - \tilde{p})}{\tilde{n}}} \quad (5.7)$$

If the lower bound is less than 0, replace it with 0. If the upper bound is greater than 1, replace it with 1.

A weighted-average method of interpolation for estimating heights from GPS measurements is evaluated. The method made “large” errors (errors whose magnitude was above a commonly accepted threshold) at 26 of the 74 sample test locations.

- a. Find a 90% confidence interval for the proportion of locations at which this method will make large errors.
- b. What sample size is needed to obtain a 95% confidence interval with width ± 0.08 ?
- c. How large a sample is needed to guarantee that the width of the 95% confidence interval will be no greater than ± 0.08 , if no preliminary sample has been taken?

a. The method made “large” errors (errors whose magnitude was above a commonly accepted threshold) at 26 of the 74 sample test locations. Find a 90% confidence interval for the proportion of locations at which this method will make large errors.

Solution

- The number of successes is $X = 26$, and the number of trials is $n = 74$. We therefore compute $\tilde{n} = 74 + 4 = 78$, $\tilde{p} = (26 + 2)/78 = 0.3590$, and $\sqrt{\tilde{p}(1 - \tilde{p})/\tilde{n}} = (0.3590)(0.6410)/78 = 0.0543$.
- For a 90% confidence interval, the value of $\alpha/2$ is 0.05, so $Z_{\alpha/2} = 1.645$.
- The 90% confidence interval is therefore $0.3590 \pm (1.645)(0.0543)$
 $= (0.270, 0.448)$.

b. what sample size is needed to obtain a 95% confidence interval with width ± 0.08 ?

Solution

- A 95% confidence interval has width $\pm 1.96 \sqrt{\tilde{p}(1 - \tilde{p}) / \tilde{n}}$, where $\tilde{n} = n + 4$.

- Therefore we determine the sample size n by solving the equation :

$$1.96 \sqrt{\tilde{p}(1 - \tilde{p}) / (n + 4)} = 0.08.$$

- $\tilde{p} = 0.3590$. Substituting this value for $\tilde{p}$ and solving, we obtain $n \approx 135$.

c. How large a sample is needed to guarantee that the width of the 95% confidence interval will be no greater than ± 0.08 , if no preliminary sample has been taken?

Solution

- Sometimes we may wish to compute a necessary sample size without having a reliable estimate $\tilde{p}$ available.
- The quantity $\tilde{p}(1 - \tilde{p})$, which determines the width of the confidence interval, is maximized for $\tilde{p} = 0.5$.
- Since the width is greatest when $\tilde{p}(1 - \tilde{p})$ is greatest, we can compute a conservative sample size estimate by substituting $\tilde{p} = 0.5$

- A 95% confidence interval has width $\pm 1.96\sqrt{\tilde{p}(1 - \tilde{p})/(n + 4)}$.
- The widest the confidence interval could be, for a sample of size n , is $\pm 1.96\sqrt{(0.5)(1 - 0.5)/(n + 4)}$, or $\pm 0.98/\sqrt{n + 4}$.
- Solving the equation $0.98/\sqrt{n + 4} = 0.08$ for n , we obtain
 $n \approx 147$.

Suppose two methods are used to compute a 95% confidence interval for a proportion based on a sample of 20 successes out of 50 trials. The traditional method gives (0.28, 0.52), while the Agresti-Coull method gives (0.30, 0.54). What is the most likely reason for the difference in intervals?

- A. The traditional method always yields a wider interval than the Agresti-Coull method.
- B. The Agresti-Coull method adjusts both the sample size and the proportion to improve performance, especially for small samples.
- C. The Agresti-Coull method is identical to the traditional method when $n > 100$.
- D. The traditional method incorporates a continuity correction to address discreteness in binomial data.

Solution: B

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Additional Concepts

i) Large-Sample Point Estimators

To estimate one of four population parameters when the sample sizes are large, use the following point estimators with the appropriate margins of error.

Parameter	Point Estimator	Margin of Error
μ	$\bar{x}$	$\pm 1.96 \left(\frac{s}{\sqrt{n}} \right)$
p	$\hat{p} = \frac{x}{n}$	$\pm 1.96 \sqrt{\frac{\hat{p}\hat{q}}{n}}$
$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$	$\pm 1.96 \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
$p_1 - p_2$	$(\hat{p}_1 - \hat{p}_2) = \left(\frac{x_1}{n_1} - \frac{x_2}{n_2} \right)$	$\pm 1.96 \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$

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Additional Concepts

ii) Large-Sample Interval Estimators

To estimate one of four population parameters when the sample sizes are large, use the following interval estimators.

Parameter	$(1 - \alpha)100\%$ Confidence Interval
μ	$\bar{x} \pm z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$
p	$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$
$\mu_1 - \mu_2$	$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
$p_1 - p_2$	$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$

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Additional Concepts

- All values in the interval are possible values for the unknown population parameter.
- Any values outside the interval are unlikely to be the value of the unknown parameter.
- To compare two population means or proportions, look for the value 0 in the confidence interval. If 0 is in the interval, it is possible that the two population means or proportions are equal, and you should not declare a difference. If 0 is not in the interval, it is unlikely that the two means or proportions are equal, and you can confidently declare a difference.

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References

“Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 6th Edition, 2024.



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MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 2: Confidence Intervals

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Mathematics for Computer Science and Engineering

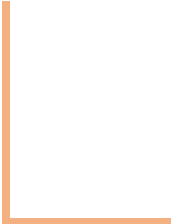
Confidence intervals for Difference between Two Means

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Topics to be covered



❖ 5.4- Confidence Intervals for the Difference between Two Means

- Sometimes we are interested in comparing the means of two populations.
- ❑ The average growth of plants fed using two different nutrients.
- ❑ The average scores for students taught with two different teaching methods.
- To make this comparison,

A random sample of size n_1 drawn from population 1 with mean μ_1 and variance σ_1^2 .

A random sample of size n_2 drawn from population 2 with mean μ_2 and variance σ_2^2 .

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Notations - Comparing Two Means

	Mean	Variance	Standard Deviation
Population 1	μ_1	σ_1^2	σ_1
Population 2	μ_2	σ_2^2	σ_2

	Sample size	Mean	Variance	Standard Deviation
Sample from Population 1	n_1	$\bar{x}_1$	s_1^2	s_1
Sample from Population 2	n_2	$\bar{x}_2$	s_2^2	s_2

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Estimating the Difference between Two Means

- We compare the two averages by making inferences about $\mu_1 - \mu_2$, the difference in the two population averages.
- If the two population averages are the same, then $\mu_1 - \mu_2 = 0$.
- The best estimate of $\mu_1 - \mu_2$ is the difference in the two sample means,

$$\bar{x}_1 - \bar{x}_2$$

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Estimating the Difference between Two Means

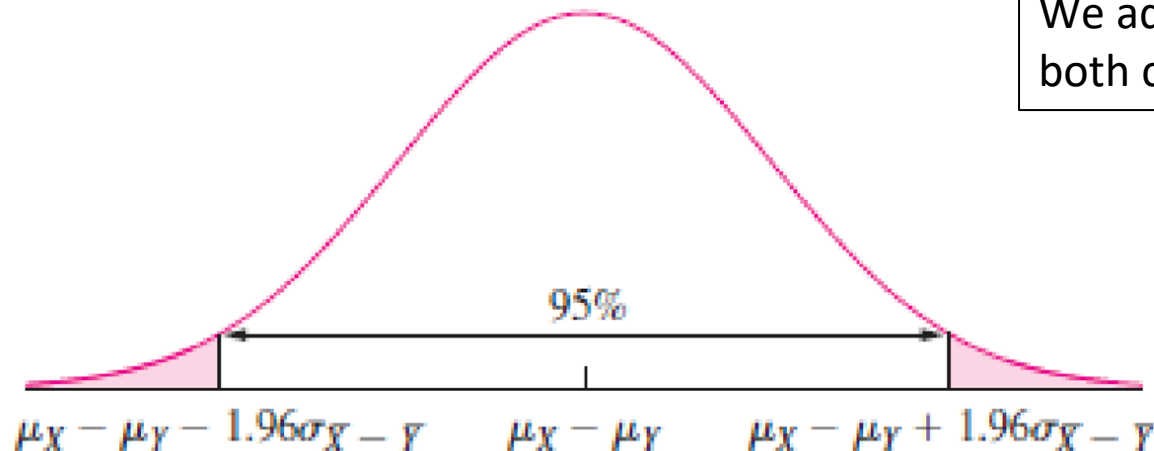
Let X and Y be independent, with $X \sim N(\mu_X, \sigma_X^2)$ and $Y \sim N(\mu_Y, \sigma_Y^2)$. Then

$$X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2) \quad (5.14)$$

$$X - Y \sim N(\mu_X - \mu_Y, \sigma_X^2 + \sigma_Y^2) \quad (5.15)$$

Note:

We add the variance for both cases.



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Estimating the Difference between Two Means

Summary

Let $X_1, \dots, X_{n_X}$ be a *large* random sample of size n_X from a population with mean μ_X and standard deviation σ_X , and let $Y_1, \dots, Y_{n_Y}$ be a *large* random sample of size n_Y from a population with mean μ_Y and standard deviation σ_Y . If the two samples are independent, then a level $100(1 - \alpha)\%$ confidence interval for $\mu_X - \mu_Y$ is

$$\bar{X} - \bar{Y} \pm z_{\alpha/2} \sqrt{\frac{\sigma_X^2}{n_X} + \frac{\sigma_Y^2}{n_Y}} \quad (5.16)$$

When the values of σ_X and σ_Y are unknown, they can be replaced with the sample standard deviations s_X and s_Y .

For large samples, point estimates and their margin of error as well as confidence intervals are based on the standard normal (z) distribution.

Point estimate for $\mu_1 - \mu_2 : \{ \bar{x}_1 - \bar{x}_2 \} \pm MoE$; where

$$\text{Margin of Error} : \pm 1.96 \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Note:

Assumption :

Both $n_1 \geq 30$ and $n_2 \geq 30$

Confidence interval for $\mu_1 - \mu_2$:

$$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Compare the average daily intake of dairy products of men and women using a 95% confidence interval.

Avg Daily Intakes	Men	Women
Sample size	50	50
Sample mean	756	762
Sample Std Dev	35	30

Compare the average daily intake of dairy products of men and women using a 95% confidence interval.

Avg Daily Intakes	Men	Women
Sample size	50	50
Sample mean	756	762
Sample Std Dev	35	30

$$(\bar{x}_1 - \bar{x}_2) \pm 1.96 \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} \Rightarrow (756 - 762) \pm 1.96 \sqrt{\frac{35^2}{50} + \frac{30^2}{50}}$$
$$\Rightarrow -6 \pm 12.78$$

$$\text{or } -18.78 < \mu_1 - \mu_2 < 6.78.$$

- Could you conclude, based on this confidence interval, that there is a difference in the average daily intake of dairy products for men and women?

$$-18.78 < \mu_1 - \mu_2 < 6.78$$

- The confidence interval **contains the value $\mu_1 - \mu_2 = 0$** . . Therefore, it is possible that **$\mu_1 = \mu_2$** .
- **You would not want to conclude that there is a difference in average daily intake of dairy products for men and women.**

Example 2

The chemical composition of soil varies with depth. The article “Sampling Soil Water in Sandy Soils: Comparative Analysis of Some Common Methods” (M. Ahmed, M. Sharma, et al., *Communications in Soil Science and Plant Analysis*, 2001: 1677–1686) describes chemical analyses of soil taken from a farm in Western Australia. Fifty specimens were each taken at depths 50 and 250 cm. At a depth of 50 cm, the average NO_3 concentration (in mg/L) was 88.5 with a standard deviation of 49.4. At a depth of 250 cm, the average concentration was 110.6 with a standard deviation of 51.5. Find a 95% confidence for the difference between the NO_3 concentrations at the two depths.

Solution

Let $X_1, \dots, X_{50}$ represent the concentrations of the 50 specimens taken at 50 cm, and let $Y_1, \dots, Y_{50}$ represent the concentrations of the 50 specimens taken at 250 cm. Then $\bar{X} = 88.5$, $\bar{Y} = 110.6$, $s_X = 49.4$, and $s_Y = 51.5$. The sample sizes are $n_X = n_Y = 50$. Both samples are large, so we can use expression (5.16). Since we want a 95% confidence interval, $z_{\alpha/2} = 1.96$. The 95% confidence interval for the difference $\mu_Y - \mu_X$ is $110.6 - 88.5 \pm 1.96\sqrt{49.4^2/50 + 51.5^2/50}$, or 22.1 ± 19.8 .

A 95% confidence interval for the difference in mean voltage output between two types of batteries is reported as (1.2, 2.8). A student interprets this result by stating:

“There is a 95% probability that the true difference in mean voltage is between 1.2 and 2.8 volts.”

Which of the following best evaluates the student’s statement?

- A. The statement is correct because the confidence level directly reflects the probability the parameter lies within the computed interval.
- B. The statement is incorrect because the confidence interval applies to sample data, not the population.
- C. The statement is incorrect; the correct interpretation is that if we repeated the experiment many times, 95% of such intervals would contain the true difference in means.
- D. The statement is correct only if the sample size exceeds 30, due to the Central Limit Theorem.

Solution: C

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References



“Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 6th Edition, 2024.



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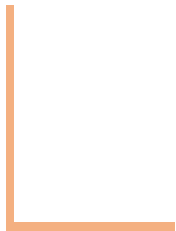
MATHEMATICS FOR COMPUTER SCIENCE ENGINEERS

Unit 2: Confidence Intervals for small samples

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Confidence intervals for mean of Small Samples



❖ 5.3 - Small-Sample Confidence Intervals for a Population Mean

- If the sample size is small, s may not be close to σ , and X may not be approximately normal.
- Point estimators remain the same
- There are small sample interval estimators/confidence intervals for
 - μ , the mean of a normal population
 - $\mu_1 - \mu_2$, the difference between two normal population means

When we take a sample from a normal population, the sample mean has a normal distribution for any sample size n , and

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

has a standard normal distribution.

But if σ is unknown, and we must use s to estimate it, the resulting statistic is not normal as s is not necessarily close to σ .

$$\frac{\bar{x} - \mu}{s / \sqrt{n}} \text{ is not normal!}$$

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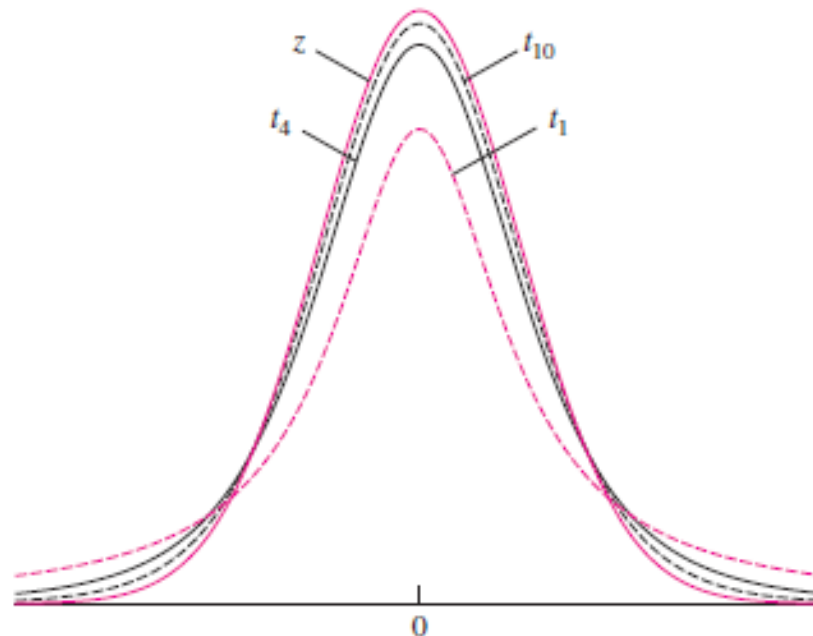
Student's t Distribution

- Fortunately, this statistic does have a sampling distribution that is well known to statisticians, called the Student's t distribution, with $n-1$ degrees of freedom, denoted by t_{n-1} .

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

- We can use this distribution to create estimation procedures for the population mean μ .

- Mound-shaped and symmetric about 0.
- More variable than z , with “heavier tails”
- Shape depends on the sample size n or the degrees of freedom, $n-1$.
- The t curves are more spread out than the normal, but the
- amount of extra spread decreases as the number of degrees of freedom increases.



- Degrees of freedom of an estimate is the **number of independent pieces of information that went into calculating the estimate.**
- **It's not quite the same as the number of items in the sample.** In order to get the df for the estimate, you have to subtract 1 from the number of items.
- Let's say you were finding the mean weight loss for a low-carb diet. You could use 4 people, giving 3 degrees of freedom ($4 - 1 = 3$), or you could use one hundred people with $df = 99$.

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Why do we subtract 1 from the number of items?

- Another way to look at degrees of freedom is that they are **the number of values that are free to vary in a data set.**
- Ex: Pick a set of numbers that have a mean (average) of 10.
- 9, 10, 11 or 8, 10, 12 or 5, 10, 15

Summary

Let $X_1, \dots, X_n$ be a *small* (e.g., $n < 30$) sample from a *normal* population with mean μ . Then the quantity

$$\frac{\bar{X} - \mu}{s/\sqrt{n}}$$

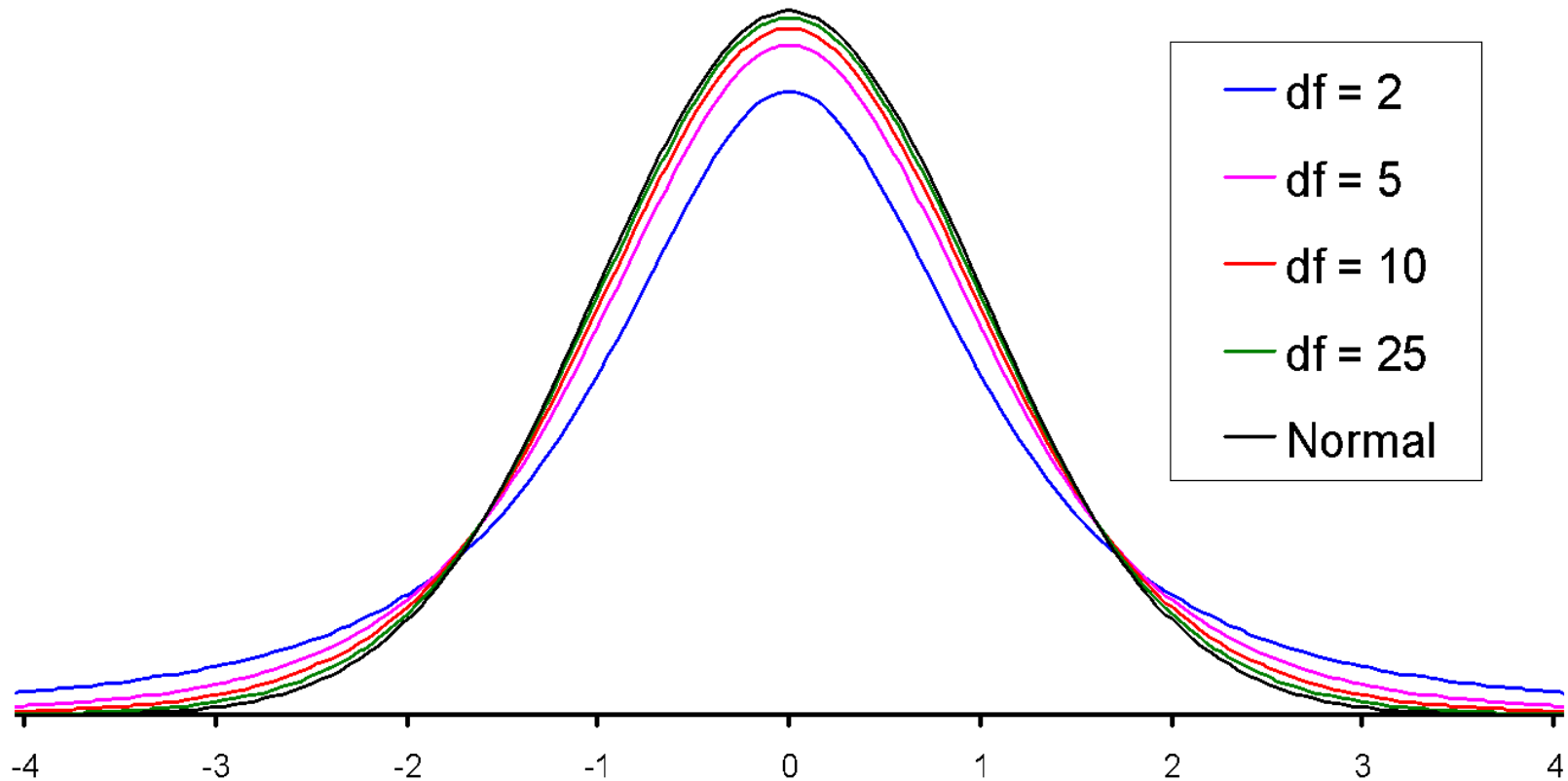
has a Student's t distribution with $n - 1$ degrees of freedom, denoted t_{n-1} .

When n is large, the distribution of the quantity $(\bar{X} - \mu)/(s/\sqrt{n})$ is very close to normal, so the normal curve can be used, rather than the Student's t .

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t distribution

Comparison of normal and t distributions



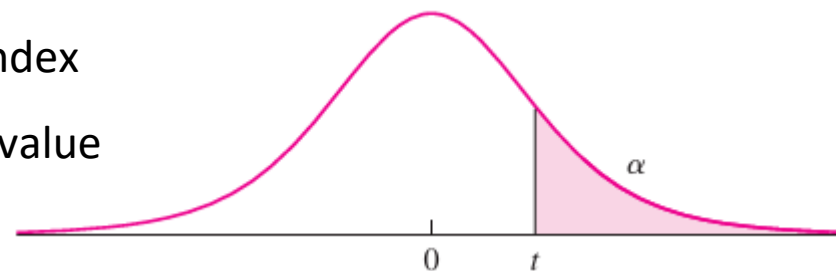
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Relationship to the NORMAL CURVE

- As the df increase, the t distribution approaches the standard normal distribution ($\mu=0.0$, $\sigma=1.0$).
- The standard normal curve is a special case of the t distribution when $df = \text{infinity}$.
- For practical purposes, the t distribution approaches the standard normal distribution relatively quickly, such that when $df=30$ the two are almost identical.
- The t-distribution is symmetric and bell-shaped, like the normal distribution, but has heavier tails, meaning that it is more prone to producing values that fall far from its mean.

Using the t-Table

Table gives the values of t that cut off certain critical values in the right tail of the t distribution. Use index df and the appropriate tail area α to find t_α , the value of t with area α to its right.



ν	α								
	0.40	0.25	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657	318.309	636.619
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.265	0.718	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169	4.144	4.587

Note:

T table on
Page 809

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Example

Let $n = 10$, find a value of t that cuts off .025 in the right tail.

So $df = n - 1 = 9$, $\alpha = 0.025$

ν	α								
	0.40	0.25	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
1	0.325	1.000	3.078	6.314	12.706	31.821	63.657	318.309	636.619
2	0.289	0.816	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.277	0.765	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.271	0.741	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.267	0.727	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.265	0.718	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.263	0.711	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.262	0.706	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.261	0.703	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.260	0.700	1.372	1.812	2.228	2.764	3.169	4.144	4.587


$$T_{0.025} = 2.262$$

Example

A random sample of size 10 is to be drawn from a normal distribution with mean 4. The Student's t statistic $t = (\bar{X} - 4)/(s/\sqrt{10})$ is to be computed. What is the probability that $t > 1.833$?

Solution

This t statistic has $10 - 1 = 9$ degrees of freedom. From the t table, $P(t > 1.833) = 0.05$. See Figure 5.10.

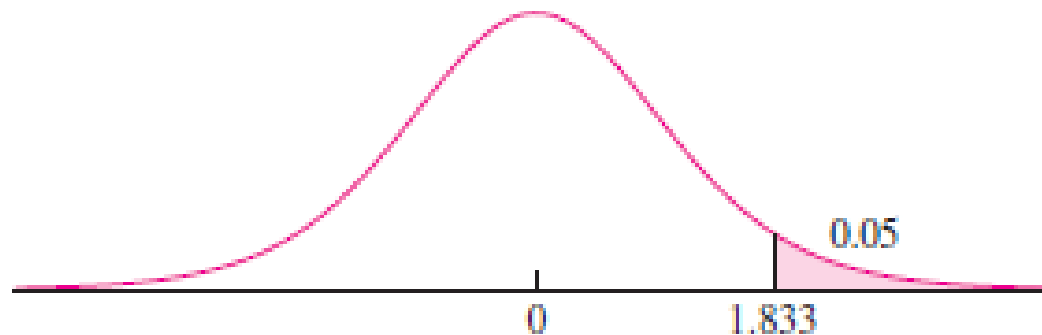


FIGURE 5.10 Solution to Example 5.15.

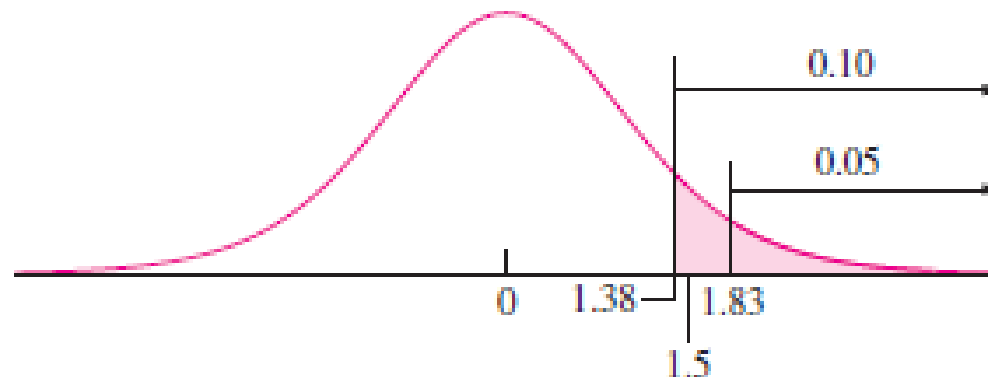
Find $P(t > 1.5)$ for the same question.

Solution

Looking across the row corresponding to 9 degrees of freedom, we see that the t table does not list the value 1.5. We find that $P(t > 1.383) = 0.10$ and $P(t > 1.833) = 0.05$. We conclude that $0.05 < P(t > 1.5) < 0.10$. See Figure 5.11. If a more precise result were required, linear interpolation could be used as follows:

$$P(t > 1.5) \approx 0.10 - \frac{1.5 - 1.383}{1.833 - 1.383}(0.10 - 0.05) = 0.0870$$

A computer package gives the answer correct to three significant digits as 0.0839.



Example

Find the value for the t_{12} distribution whose upper-tail probability is 0.025.

	α				
ν	0.40	0.25	0.10	0.05	0.025
1	0.325	1.000	3.078	6.314	12.706
2	0.289	0.816	1.886	2.920	4.303
3	0.277	0.765	1.638	2.353	3.182
4	0.271	0.741	1.533	2.132	2.776
5	0.267	0.727	1.476	2.015	2.571
6	0.265	0.718	1.440	1.943	2.447
7	0.263	0.711	1.415	1.895	2.365
8	0.262	0.706	1.397	1.860	2.306
9	0.261	0.703	1.383	1.833	2.262
10	0.260	0.700	1.372	1.812	2.228
11	0.260	0.697	1.363	1.796	2.201
12	0.259	0.695	1.356	1.782	2.179
13	0.259	0.694	1.350	1.771	2.160
14	0.258	0.692	1.345	1.761	2.145

Find the value for the t_{14} distribution whose lower-tail probability is 0.01.

Solution

Look down the column headed “0.01” to the row corresponding to 14 degrees of freedom. The value for t_{14} is 2.624. This value cuts off an area, or probability, of 1% in the upper tail. The value whose lower-tail probability is 1% is -2.624 .



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Small Sample Confidence Interval for Population Mean

Summary

Let $X_1, \dots, X_n$ be a *small* random sample from a *normal* population with mean μ . Then a level $100(1 - \alpha)\%$ confidence interval for μ is

$$\bar{X} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}} \quad (5.9)$$

Ten randomly selected students were each asked to list how many hours of television they watched per month. The results are

82	66	90	84	75
88	80	94	110	91

Find a 90% confidence interval for the true mean number of hours of television watched per month by students.

Calculating the sample mean and standard deviation we have $n = 10$, $\bar{x} = 86$, and $s = 11.842$. We find the critical t value of 1.833 by looking on the t table in the row corresponding to $df = 9$, in the column with label $t_{.050}$. The 90% confidence interval for μ is

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} = 86 \pm (1.833) \frac{11.842}{\sqrt{10}} = 86 \pm 6.86$$
$$(79.14, 92.86)$$

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Is the Student's t Distribution Appropriate?

- **Don't Use the Student's t Statistic If the Sample Contains Outliers**

- A reasonable way to proceed is to **construct a boxplot or dotplot of**

the sample. If these plots **do not reveal a strong asymmetry or any outliers, then in most cases the Student's t distribution will be reliable.**

- **Use z, Not t, If σ Is Known**

- Occasionally a small sample may be taken from a normal population whose standard deviation σ is known.

- In these cases, we do not use the Student's t curve, because we are not approximating σ with s . Instead, we use the z table.

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Example



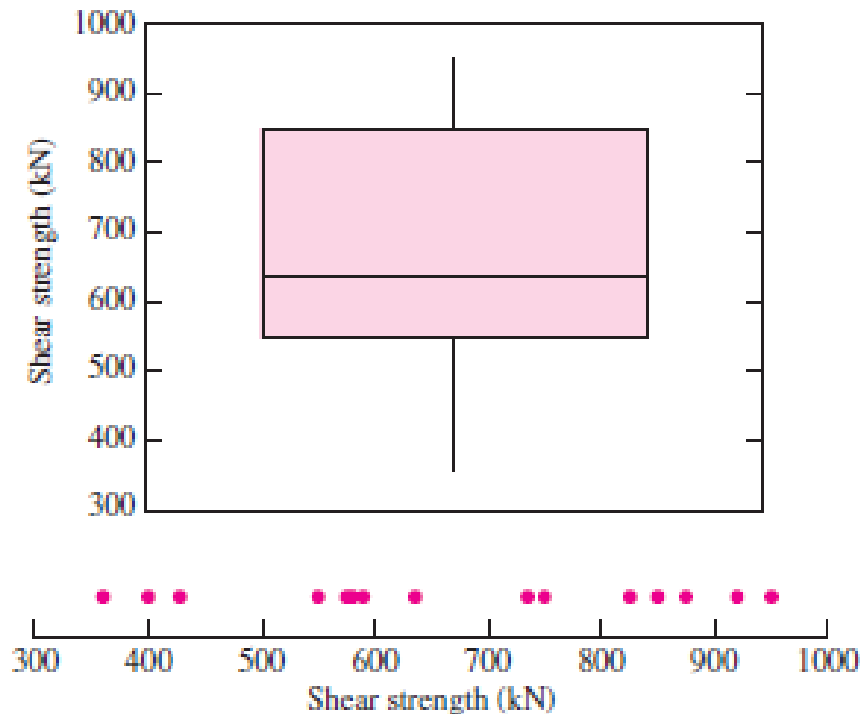
An article presents measurements of the nominal shear strength (in kN) for a **sample of 15** prestressed concrete beams. The results are:

580 400 428 825 850 875 920 550
575 750 636 360 590 735 950

Is it appropriate to use the Student's t statistic to **construct a 99% confidence interval** for the mean shear strength? If so, construct the confidence interval. If not, explain why not.

Solution

To determine whether the Student's t statistic is appropriate, we will make a boxplot and a dotplot of the sample. These are shown in the following figure.



Solution:

No outliers and no asymmetry

$\bar{X} = 668.27$ and $s = 192.089$.
with $n = 15$ and $\alpha/2 = 0.005$. From the t table with 14 degrees of freedom, we find $t_{14, 0.005} = 2.977$.

The 99% confidence interval is $668.27 \pm (2.977)(192.089)/\sqrt{15}$,
Or (520.62, 815.92).

Example

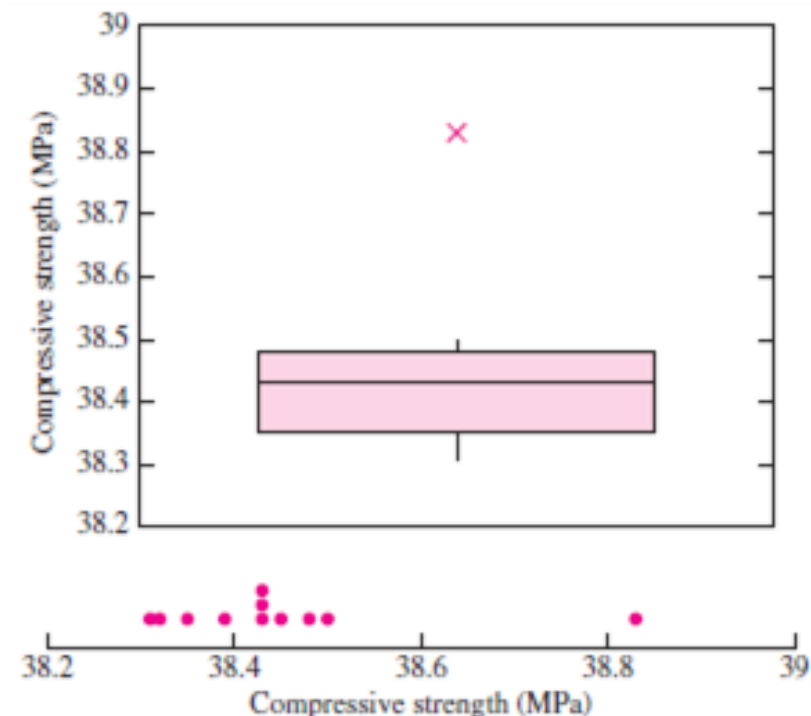
In the article referred to in Example 5.19, cylindrical compressive strength (in MPa) was measured for 11 beams. The results were

38.43 38.43 38.39 38.83 38.45 38.35 38.43 38.31 38.32 38.48 38.50

Is it appropriate to use the Student's t statistic to construct a 95% confidence interval for the mean cylindrical compressive strength? If so, construct the confidence interval. If not, explain why not.

Solution:

There is an outlier in this sample. The Student's t statistic should not be used.



An engineer reads a report that states that a sample of 11 concrete beams had an average compressive strength of 38.45 MPa with standard deviation 0.14 MPa. Should the t curve be used to find a confidence interval for the mean compressive strength?

Solution

No. The problem is that there is no way of knowing whether the measurements came from a normal population. For example, if the measurements contained an outlier (as in Example 5.20), the confidence interval would be invalid.

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To Remember

- It is important to remember that **when the sample size is small, the population must be approximately normal, whether or not the standard deviation is known.**

The Student's t distribution can be used to compute one-sided confidence intervals. The formulas are analogous to those used with large samples.

Let $X_1, \dots, X_n$ be a *small* random sample from a *normal* population with mean μ . Then a level $100(1 - \alpha)\%$ upper confidence bound for μ is

$$\bar{X} + t_{n-1, \alpha} \frac{s}{\sqrt{n}} \quad (5.10)$$

and a level $100(1 - \alpha)\%$ lower confidence bound for μ is

$$\bar{X} - t_{n-1, \alpha} \frac{s}{\sqrt{n}} \quad (5.11)$$

Summary

Let $X_1, \dots, X_n$ be a random sample (of any size) from a *normal* population with mean μ . If the standard deviation σ is known, then a level $100(1 - \alpha)\%$ confidence interval for μ is

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad (5.12)$$

Summary

Let X be a single value sampled from a *normal* population with mean μ . If the standard deviation σ is known, then a level $100(1 - \alpha)\%$ confidence interval for μ is

$$X \pm z_{\alpha/2} \sigma \quad (5.13)$$

The article “Direct Strut-and-Tie Model for Prestressed Deep Beams” (K. Tan, K. Tong, and C. Tang, *Journal of Structural Engineering*, 2001:1076–1084) presents measurements of the nominal shear strength (in kN) for a sample of 15 prestressed concrete beams. The results are

580	400	428	825	850	875	920	550
575	750	636	360	590	735	950	

Assume that on the basis of a very large number of previous measurements of other beams, the population of shear strengths is known to be approximately normal, with standard deviation $\sigma = 180.0$ kN. Find a 99% confidence interval for the mean shear strength.

Solution

We compute $\bar{X} = 668.27$. We do not need to compute s , because we know the population standard deviation σ . Since we want a 99% confidence interval, $\alpha/2 = 0.005$. Because we know σ , we use $z_{\alpha/2} = z_{.005}$, rather than a Student's t value, to compute the confidence interval. From the z table, we obtain $z_{.005} = 2.58$. The confidence interval is $668.27 \pm (2.58)(180.0)/\sqrt{15}$, or (548.36, 788.18).

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References

- “Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 6th Edition, 2015.



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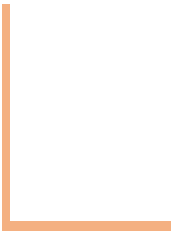
Unit 2: Confidence Intervals: Difference Between Two Proportions

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Mathematics for Computer Science and Engineering

Confidence Intervals: Difference Between Two Proportions



❖ 5.5 Confidence Intervals for the Difference Between Two Proportions

- Used to assess if proportions differ between two populations.
- Common in comparing treatments, machines, or populations.
- Example: Defect rates from two production lines.

- In a Bernoulli population, the mean is equal to the success probability p , which is the proportion of successes in the population.
- When independent trials are performed from each of two Bernoulli populations, we can use methods similar to that of difference between two means to find a confidence interval for the difference between the two success probabilities.

Let X be the number of successes in n_X independent Bernoulli trials with success probability p_X , and let Y be the number of successes in n_Y independent Bernoulli trials with success probability p_Y , so that $X \sim \text{Bin}(n_X, p_X)$ and $Y \sim \text{Bin}(n_Y, p_Y)$. Define $\tilde{n}_X = n_X + 2$, $\tilde{n}_Y = n_Y + 2$, $\tilde{p}_X = (X + 1)/\tilde{n}_X$, and $\tilde{p}_Y = (Y + 1)/\tilde{n}_Y$.

Then a level $100(1 - \alpha)\%$ confidence interval for the difference $p_X - p_Y$ is

$$\tilde{p}_X - \tilde{p}_Y \pm z_{\alpha/2} \sqrt{\frac{\tilde{p}_X(1 - \tilde{p}_X)}{\tilde{n}_X} + \frac{\tilde{p}_Y(1 - \tilde{p}_Y)}{\tilde{n}_Y}} \quad (5.20)$$

If the lower limit of the confidence interval is less than -1 , replace it with -1 . If the upper limit of the confidence interval is greater than 1 , replace it with 1 .

- Eighteen of 60 light trucks produced on assembly line A had a defect in the steering mechanism, which needed to be repaired before shipment.
- Only 16 of 90 trucks produced on assembly line B had this defect.
- Assume that these trucks can be considered to be two independent simple random samples from the trucks manufactured on the two assembly lines.
- We wish to find a 95% confidence interval for the difference between the proportions of trucks with this defect on the two assembly lines.

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solution

we define $\tilde{n}_X = n_X + 2$, $\tilde{n}_Y = n_Y + 2$, $\tilde{p}_X = (X + 1)/\tilde{n}_X$, and $\tilde{p}_Y = (Y + 1)/\tilde{n}_Y$.

The 95% confidence interval is $\tilde{p}_X - \tilde{p}_Y \pm z_{\alpha/2} \sqrt{\tilde{p}_X(1 - \tilde{p}_X)/\tilde{n}_X + \tilde{p}_Y(1 - \tilde{p}_Y)/\tilde{n}_Y}$. In this example, $\tilde{n}_X = 62$, $\tilde{n}_Y = 92$, $\tilde{p}_X = 19/62 = 0.3065$, and $\tilde{p}_Y = 17/92 = 0.1848$. We thus obtain $0.3065 - 0.1848 \pm 0.1395$, or $(-0.0178, 0.2612)$.

The article “Recent Findings on the Prevalence of E-Cigarette Use Among Adults in the United States” (F. Wilson and Y. Wang, American Journal of Preventive Medicine, 2017:385–390) surveyed adults aged 18–64 to ask whether they had ever used e-cigarettes. In a sample of 16,611 men, 2350 said they had used e-cigarettes, and in a sample of 17,745 women, 1929 said they had used them. Find a 95% confidence interval for the difference between the proportions of men and women who have used e-cigarettes.

The number of successes among men is $X = 2350$ and the number of successes among women is $Y = 1929$. The numbers of trials are $n_X = 16,611$ and $n_Y = 17,745$. We compute

$\tilde{n}_X = 16,613$, $\tilde{n}_Y = 17,747$, $\tilde{p}_X = (2350 + 1)/16,613 = 0.14152$, and $\tilde{p}_Y = (1929 + 1)/17,747 = 0.10875$. The value of $z_{\alpha/2}$ is 1.96. The 95% confidence interval is $0.14152 - 0.10875$

$\pm 1.96 \sqrt{(0.14152)(0.85848)/16,613 + (0.10875)(0.89125)/17,747}$, or 0.0328 ± 0.0070 .

- Many people use the traditional method for computing confidence intervals for the difference between proportions. This method uses the sample proportions and the actual sample sizes and .
- The traditional method gives results very similar to those of the modern method previously described for large or moderately large sample sizes.
- For small sample sizes, however, the traditional confidence interval fails to achieve its coverage probability; in other words, level confidence intervals computed by the traditional method cover the true value less than of the time.

The Traditional Method for Computing Confidence Intervals for the Difference Between Proportions (widely used but not recommended)

Let $\hat{p}_X$ be the proportion of successes in a *large* number n_X of independent Bernoulli trials with success probability p_X , and let $\hat{p}_Y$ be the proportion of successes in a *large* number n_Y of independent Bernoulli trials with success probability p_Y . Then the traditional level $100(1 - \alpha)\%$ confidence interval for $p_X - p_Y$ is

$$\hat{p}_X - \hat{p}_Y \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_X(1 - \hat{p}_X)}{n_X} + \frac{\hat{p}_Y(1 - \hat{p}_Y)}{n_Y}} \quad (5.21)$$

This method cannot be used unless both samples contain at least 10 successes and 10 failures.

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References

- “Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 6th Edition, 2025.



Thank you

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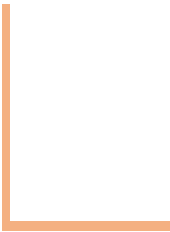
Confidence Intervals: Paired data

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Topics to be covered

❖ 5.7 - Confidence Intervals with Paired Data

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Confidence intervals for paired data

- We have seen how to compare two independent samples
- Now **we will see how to compare two samples that are paired** ?
- In other words **the two samples are not independent, Y_1 and Y_2 are linked in some way, usually by a direct relationship** ?
- *For example,*
 - measure the weight of subjects before and after a six month diet.
 - A tire manufacturer wishes to compare the tread wear of tires made of a new material with that of tires made of a conventional material.

- A **paired t-test** is used to compare two population means where you have two samples in which observations in one sample can be paired with observations in the other sample.
- Examples of where this might occur are:
 - **Before-and-after observations on the same subjects**
 - (e.g. students' diagnostic test results before and after a particular module or course).
- A comparison of two different methods of measurement or two different treatments where the measurements/treatments are applied to the same subjects.

Example

Let x = test score before the module, y = test score after the module .To find that the true mean difference is zero, the procedure is as follows:

1. Calculate the difference ($d_i = y_i - x_i$) between the two observations on each pair, making sure you distinguish between positive and negative differences.
2. Calculate the mean difference, $\bar{d}$.
3. Calculate the standard deviation of the differences, s_d , and use this to calculate the standard error of the mean difference, $SE(\bar{d}) = \frac{s_d}{\sqrt{n}}$
4. Calculate the t-statistic, which is given by $T = \frac{\bar{d}}{SE(d)}$. Under the null hypothesis, this statistic follows a t-distribution with $n - 1$ degrees of freedom.
5. Use tables of the t-distribution to compare your value for T to the t_{n-1} distribution. This will give the p-value for the paired t-test.

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To Remember

For this test to be valid the differences only need to be approximately normally distributed. Therefore, it would not be advisable to use a paired t-test where there were any extreme outliers.

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Example

A tire manufacturer wishes to **compare the tread wear of tires made of a new material with that of tires made of a conventional material**. One tire of each type is placed on **each front wheel of each of 10 front-wheel-drive automobiles**. The choice as to which **type of tire goes on the right wheel and which goes on the left is made with the flip of a coin**. Each car is driven for 40,000 miles, then the tires are removed, and the depth of the tread on each is measured.

We wish to find a 95% confidence interval for the mean difference in tread wear between old and new materials in a way that takes advantage of the reduced variability produced by the paired design.

	Car									
	1	2	3	4	5	6	7	8	9	10
New material	4.35	5.00	4.21	5.03	5.71	4.61	4.70	6.03	3.80	4.70
Old material	4.19	4.62	4.04	4.72	5.52	4.26	4.27	6.24	3.46	4.50
Difference	0.16	0.38	0.17	0.31	0.19	0.35	0.43	-0.21	0.34	0.20

To put this into statistical notation, let $(X_1, Y_1), \dots, (X_{10}, Y_{10})$ be the pairs.
(X_i $\rightarrow$ tread on the tire made from new material on the i^{th} car) and
(Y_i $\rightarrow$ tread on the tire made from old material on the i^{th} car.)

Let $D_i = X_i - Y_i$ represent the difference between treads for tires on the i^{th} car. Let μ_X and μ_Y represent the population means for X and Y , respectively.

We wish to find a 95% confidence interval for the difference $\mu_X - \mu_Y$.

Let μ_D represent the population mean of the differences.

Then $\mu_D = \mu_X - \mu_Y$.

It follows that a confidence interval for μ_D will also be a confidence interval for $\mu_X - \mu_Y$.

Since the sample $D_1, \dots, D_{10}$ is a random sample from a population with mean μ_D , we can use one-sample methods to find confidence intervals for μ_D .

In this example, since the sample size is small, we use the Student's t method.

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Example

$$\overline{D} = 0.232 \quad s_D = 0.183$$

The sample size is 10, so there are nine degrees of freedom. The appropriate t value is $t_{9,.025} = 2.262$.

The confidence interval

therefore $0.232 \pm (2.262)(0.183)/\sqrt{10}$, or (0.101, 0.363).

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Confidence interval for the true mean difference

Summary

Let $D_1, \dots, D_n$ be a *small* random sample ($n \leq 30$) of differences of pairs. If the population of differences is approximately normal, then a level $100(1 - \alpha)\%$ confidence interval for the mean difference μ_D is given by

$$\bar{D} \pm t_{n-1, \alpha/2} \frac{s_D}{\sqrt{n}} \quad (5.24)$$

where s_D is the sample standard deviation of $D_1, \dots, D_n$. Note that this interval is the same as that given by expression (5.9).

If the sample size is large, a level $100(1 - \alpha)\%$ confidence interval for the mean difference μ_D is given by

$$\bar{D} \pm z_{\alpha/2} \sigma_{\bar{D}} \quad (5.25)$$

In practice $\sigma_{\bar{D}}$ is approximated with $s_D/\sqrt{n}$. Note that this interval is the same as that given by expression (5.1).

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Factors Affecting Margin of Error

Sample Size (n)	Larger n $\rightarrow$ Smaller MOE Smaller n $\rightarrow$ Larger MOE	More data reduces variability, making estimates more precise.	n = 100 $\rightarrow$ MOE $\approx$ 5% n = 1000 $\rightarrow$ MOE $\approx$ 1.6%
Standard deviation (variability)	Higher variability $\rightarrow$ Larger MOE Lower variability $\rightarrow$ Smaller MOE	When data are spread out, estimates fluctuate more from sample to sample.	$\sigma = 20$ (test scores) $\rightarrow$ wide MOE $\sigma = 5 \rightarrow$ narrow MOE
Confidence Level	Higher confidence $\rightarrow$ Larger MOE Lower confidence $\rightarrow$ Smaller MOE	To be more certain, interval must be widened (depends on z score or t score).	90% CI $\rightarrow$ z score $\approx$ 1.645 95% CI $\rightarrow$ z score $\approx$ 1.96 99% CI $\rightarrow$ z score $\approx$ 2.576

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References

- “Statistics for Engineers and Scientists”, William Navidi, McGraw Hill Education, India, 4th Edition, 2015.



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